

TACHS Full-Length Practice Exam 06

Reading · Written Expression · Mathematics · Ability

This printable contains 200 multiple-choice questions arranged in four parts, followed by the validated answer keys and explanations supplied with each part.

Recommended section times: Reading 40 minutes · Written Expression 40 minutes · Mathematics 45 minutes · Ability 35 minutes

Exam Part A

Filename: tachs_full_exam_06_part_a.txt

Exam Family: TACHS

Exam Number: 6

Exam Part: A

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R6-S01

Section: Reading

Stimulus Type: Literary Passage

Title: Seven Seconds of Silence

Text:

Eli had volunteered to operate the soundboard for the school's weekly radio program because the job seemed safely invisible. He adjusted music levels, watched the clock, and signaled the student hosts through a glass window. Speaking belonged to someone else.

On Friday, a local baker was scheduled for an interview about reducing food waste. Two minutes before airtime, the host sent a message saying her bus had stalled. The second host was already reading announcements in the studio. Eli could either cancel the interview or conduct it himself.

When the baker arrived, Eli opened the microphone and asked his first prepared question. Then he glanced at the next line too early and missed part of the response. Seven seconds of silence filled his headphones. Instead of rushing ahead, he asked, "What made that first experiment work?" The baker described turning day-old bread into crisp toppings, and the conversation began to flow.

After the broadcast, Eli apologized for the pause. The baker laughed. "I thought you were listening." Eli replayed the recording. The silence no longer sounded like an empty space. It sounded like the moment he stopped hiding behind his list and became curious about the answer.

STIMULUS END

TACHS-R6-001

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S01

Prompt:

Why is the scheduled host unable to conduct the interview?

- A) The soundboard stops working.
- B) She has lost the question list.
- C) The baker arrives late.
- D) Her bus has stalled.

TACHS-R6-002

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S01

Prompt:

Why does Eli ask what made the first experiment work?

- A) He has become interested in what the baker just said.
- B) He wants to change the interview topic completely.
- C) He needs time to repair the studio microphone.
- D) He wants the baker to repeat the prepared introduction.

TACHS-R6-003

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S01

Prompt:

Which statement best expresses a central idea of the passage?

- A) Public speaking always requires memorizing every question.
- B) Listening closely can turn an uncomfortable pause into meaningful conversation.
- C) Technical jobs are more important than speaking roles.
- D) Unexpected problems are best solved by canceling a plan.

TACHS-R6-004

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S01

Prompt:

As used in the final paragraph, “hiding” most nearly means

- A) avoiding active participation by relying on the list.
- B) covering the soundboard with equipment.
- C) leaving the studio without permission.
- D) placing an object where it cannot be found.

STIMULUS START

Stimulus ID: TACHS-R6-S02

Section: Reading

Stimulus Type: Informational Passage

Title: A Library of Scents

Text:

Museums preserve objects that can be seen and touched, but some researchers also collect scents. Smell is part of the experience of places and materials, yet it is difficult to store. An old book, a wooden workshop, or a historic soap may lose its characteristic odor as its ingredients change over time.

To document a scent, researchers may place an object in a sealed container with a material that traps airborne molecules. Laboratory instruments then separate and identify many of those molecules. A written “odor profile” records both the chemical findings and descriptions supplied by human volunteers, such as smoky, grassy, or sweet.

Re-creating the scent is a separate challenge. A list of molecules does not automatically reveal the exact amounts needed, and people may describe the same odor differently. Researchers therefore compare several trial mixtures with the original object. They also avoid exposing fragile artifacts to substances that might damage them.

Scent archives can support conservation and education. If a material's odor changes, that shift may provide evidence of decay. A carefully reconstructed scent can also help visitors imagine a setting without touching its original objects. Such collections do not replace visual records; they preserve another layer of information that would otherwise disappear.

STIMULUS END

TACHS-R6-005

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S02

Prompt:

What is recorded in an odor profile?

- A) Only the age and weight of the object
- B) Chemical findings and human descriptions
- C) A photograph of every trial mixture
- D) Instructions for touching the artifact

TACHS-R6-006

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S02

Prompt:

Why does the author distinguish documenting a scent from re-creating it?

- A) To show that identifying components does not determine a complete mixture
- B) To argue that human descriptions have no scientific value
- C) To prove that old objects should always be sealed
- D) To suggest that instruments can reproduce any odor immediately

TACHS-R6-007

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R6-S02

Prompt:

How is the passage mainly organized?

- A) It introduces a preservation problem, explains methods and challenges, and describes possible uses.
- B) It lists scents from weakest to strongest.
- C) It presents a debate between researchers and visitors.

D) It compares two museums and declares one superior.

TACHS-R6-008

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S02

Prompt:

Which finding would most likely concern a conservator?

- A) Visitors use different words for the same pleasant scent.
- B) A wooden artifact develops an odor associated with chemical breakdown.
- C) A replica has a stronger color than an original object.
- D) A sealed container occupies little storage space.

STIMULUS START

Stimulus ID: TACHS-R6-S03

Section: Reading

Stimulus Type: Humanities Passage

Title: The Margin as a Meeting Place

Text:

Many readers are taught to keep library books free of marks, and for good reason. Yet in books people own, handwritten notes can reveal an active conversation between reader and text. A question beside a puzzling paragraph, a line connecting two ideas, or a brief disagreement records thought at the moment it occurs.

Marginal notes also allow a reader to meet an earlier version of oneself. A comment that once seemed insightful may later appear incomplete. Rather than proving the first reading was useless, the contrast shows that understanding has changed. The page becomes a record of revision, not just of information received.

Some notes face outward as well. Families sometimes pass books from one generation to another, and scholars study annotations in writers' personal libraries. These marks can reveal which passages attracted attention or how an idea developed. Still, a note is evidence of one response, not a final explanation of what a passage means.

Digital reading tools can preserve comments without placing ink on paper. Whether written with pencil or typed on a screen, useful annotation does more than decorate a page. It identifies a connection, asks a focused question, or tests an interpretation. In this way, the margin becomes not an escape from the text but a place to engage with it.

STIMULUS END

TACHS-R6-009

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S03

Prompt:

According to the passage, what can scholars learn from annotations in a writer's books?

- A) Whether all readers understood a passage identically
- B) The exact price paid for every book
- C) Which passages drew attention or how an idea developed

D) How quickly the writer read each page

TACHS-R6-010

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S03

Prompt:

As used in the second paragraph, “contrast” most nearly means

- A) decoration.
- B) difference.
- C) prediction.
- D) agreement.

TACHS-R6-011

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S03

Prompt:

Which statement best summarizes the passage?

- A) Writing in every book is harmless and should be required.
- B) Thoughtful annotations record and support a reader’s changing engagement with a text.
- C) Older interpretations are always less accurate than newer ones.
- D) Digital comments are more useful than handwritten notes.

TACHS-R6-012

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S03

Prompt:

Why does the author warn that a note is “not a final explanation”?

- A) A reader’s response provides evidence but remains one interpretation.
- B) Most notes are too faded for scholars to read.
- C) Only the book’s author may explain a passage.
- D) Annotations always contradict the printed text.

TACHS-R6-013

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S03

Prompt:

The author’s tone toward thoughtful annotation is best described as

- A) suspicious and dismissive.
- B) amused but unconvinced.
- C) neutral and uninterested.
- D) appreciative but qualified.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE6-S01

Section: Written Expression

Stimulus Type: Revision Passage

Title: Starting a Seed Library

Text:

[1] Our community center plans to open a seed library where gardeners can take small packets of seeds and later contribute seeds they collect. [2] Before accepting donations, volunteers will provide instructions for labeling each packet with the plant name and collection year. [3] Clear labels will help users choose seeds and help volunteers remove packets that are too old. [4] Gardening is a hobby enjoyed in many countries. [5] The seeds will be stored in dry containers away from direct sunlight. [6] During monthly workshops, experienced gardeners can demonstrate how to collect mature seeds without mixing plant varieties. [7] With careful organization, the seed library can help neighbors share both materials and knowledge.

STIMULUS END

TACHS-WE6-001

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S01

Prompt:

Which sentence should be removed because it does not support the passage's main idea?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 3
- D) Sentence 6

TACHS-WE6-002

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S01

Prompt:

Which revision most effectively combines sentences 2 and 3?

- A) Because clear labels help users choose seeds and volunteers remove old packets, volunteers will explain how to record the plant name and collection year.
- B) Volunteers will provide labeling instructions, clear labels helping users choose seeds and volunteers remove old packets.
- C) Volunteers provide instructions, and labeling packets, which helps with seeds that are old.
- D) Clear labels for plant names and years, volunteers will tell donors before accepting them.

TACHS-WE6-003

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S01

Prompt:

Which revision of sentence 5 is most precise?

- A) The stuff will be put somewhere good for it.
- B) The packets will remain in airtight containers in a cool, dry cabinet away from sunlight.
- C) They will store them in containers and things away from there.
- D) Seeds are seeds that need storage in a place.

TACHS-WE6-004

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S01

Prompt:

Where would the following sentence fit best? "A simple checkout list will record which seed types are taken most often."

- A) Before sentence 1
- B) After sentence 1
- C) After sentence 4
- D) After sentence 7

STIMULUS START

Stimulus ID: TACHS-WE6-S02

Section: Written Expression

Stimulus Type: Revision Passage

Title: Measuring Nighttime Light

Text:

[1] Our class investigated how much artificial light reaches different parts of the school grounds at night. [2] Working in teams, light readings were recorded at six locations. [3] Each team held the same meter at waist height and waited ten seconds before recording a value. [4] We also noted nearby lamps, windows, and reflective surfaces. [5] The brightest reading came from the walkway beside the gym, where three uncovered lamps faced outward. [6] The garden behind the library was dimmer than every other location. [7] Repeating the measurements on a second evening would help us determine whether the pattern is consistent.

STIMULUS END

TACHS-WE6-005

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S02

Prompt:

Which revision of sentence 2 corrects the misplaced modifier?

- A) Working in teams, six locations recorded the students' light readings.
- B) The light readings, working in teams, were at six locations.
- C) Working in teams, the students recorded light readings at six locations.
- D) At six locations were readings recorded by teams working.

TACHS-WE6-006

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S02

Prompt:

Which transition should be added to the beginning of sentence 4?

- A) For example,
- B) In addition,
- C) Instead,
- D) Nevertheless,

TACHS-WE6-007

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S02

Prompt:

Which revision corrects the error in sentence 6?

- A) The garden behind the library was dimmer then every other location.
- B) The garden behind the library was more dim then every other location.
- C) The garden behind the library was dimmer than every other location.
- D) The garden behind the library was dimmest than every other location.

TACHS-WE6-008

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S02

Prompt:

Which revision of sentence 7 is most concise without changing its meaning?

- A) Repeating again on another second evening would help with seeing the same consistent pattern.
- B) If we repeated measurements again for a second time, it might possibly help us determine consistency.
- C) A second evening of measurements would show whether the pattern remains consistent.
- D) The measurements are something we could do and repeat on an evening that is second.

TACHS-WE6-009

Section: Written Expression

Type: mcq

Category: Written Expression
Skill: Grammar and Usage
Stimulus ID: None
Prompt:

Which sentence is written correctly?

- A) Every one of the windows were closed before the storm.
- B) One of the maps shows a trail beside the river.
- C) Neither the coach nor the players was ready to leave.
- D) The scissors is stored in the top drawer.

TACHS-WE6-010

Section: Written Expression
Type: mcq
Category: Written Expression
Skill: Sentence Structure
Stimulus ID: None
Prompt:

Which choice is a complete sentence?

- A) Although the rain stopped before noon.
- B) Because the game had already been delayed.
- C) While waiting for the field to dry.
- D) The rain stopped before noon, and the field dried quickly.

TACHS-WE6-011

Section: Written Expression
Type: mcq
Category: Written Expression
Skill: Grammar and Usage
Stimulus ID: None
Prompt:

Which sentence uses the pronoun correctly?

- A) The teacher gave Marcus and I the keys.
- B) Her and Lena prepared the display.
- C) Please send the completed form to him and me.
- D) The final decision belongs to they.

TACHS-WE6-012

Section: Written Expression
Type: mcq
Category: Written Expression
Skill: Error Identification
Stimulus ID: None
Prompt:

Which numbered line contains an error in standard written US English?

- Line 1: The robotics club meet every Thursday.
- Line 2: Its members design machines for local contests.
- Line 3: A science teacher supervises each meeting.

Line 4: Parents sometimes help gather materials.

- A) Line 4
- B) Line 2
- C) Line 3
- D) Line 1

TACHS-WE6-013

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: We carried the empty boxes into the storage room.

Line 2: Maya stacked the smallest boxes on a shelf.

Line 3: The larger boxes were placed beside the door.

Line 4: Later, the custodian check the arrangement.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M6-001

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $7/8 \div 14/15$?

- A) $16/15$
- B) $49/120$
- C) $15/16$
- D) $21/22$

TACHS-M6-002

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $5x - 9 = 36$.

- A) 7

- B) 8
- C) 11
- D) 9

TACHS-M6-003

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The numbers of pages read are 18, 22, 22, 25, and 33. What is the median?

- A) 22
- B) 20
- C) 18
- D) 25

TACHS-M6-004

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A circle has a radius of 7 centimeters. Using $\pi = 22/7$, what is its circumference?

- A) 22 centimeters
- B) 28 centimeters
- C) 44 centimeters
- D) 154 centimeters

TACHS-M6-005

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A container holds 3.5 liters. How many milliliters is this?

- A) 35 milliliters
- B) 3,500 milliliters
- C) 3,050 milliliters
- D) 350 milliliters

TACHS-M6-006

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which number is a common multiple of 12 and 18?

- A) 24
- B) 54
- C) 48
- D) 36

TACHS-M6-007

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

The sequence begins 5, 11, 23, 47. Which rule produces each term after the first?

- A) Add 6.
- B) Multiply by 2.
- C) Add consecutive even numbers.
- D) Multiply by 2, then add 1.

TACHS-M6-008

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A spinner has eight equal sections numbered 1 through 8. What is the probability of landing on a factor of 8?

- A) $\frac{1}{4}$
- B) $\frac{3}{8}$
- C) $\frac{1}{2}$
- D) $\frac{5}{8}$

TACHS-M6-009

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

Two angles of a triangle measure 38° and 67° . What is the measure of the third angle?

- A) 65°
- B) 70°
- C) 85°
- D) 75°

TACHS-M6-010

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A runner completes 1.2 kilometers in each lap. How far does the runner travel in 5 laps?

- A) 5 kilometers
- B) 6 kilometers
- C) 5.2 kilometers
- D) 6.2 kilometers

TACHS-M6-011

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A jacket originally costs \$80 and is discounted 15%. What is the sale price?

- A) \$72
- B) \$12
- C) \$65
- D) \$68

TACHS-M6-012

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Which inequality represents “three less than twice n is at least 15”?

- A) $3 - 2n \geq 15$
- B) $2(n - 3) \geq 15$
- C) $2n - 3 \geq 15$
- D) $2n + 3 \leq 15$

SECTION 4 - ABILITY**TACHS-A6-001**

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the right cell adds one open shape of the same kind.

Row 1: ○ ○ | ○ ○ ○

Row 2: △ △ △ | ?

A) △ △ △ △ △

B) △ △

C) ▲ ▲ ▲ ▲

D) △ △ △ △

TACHS-A6-002

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Down each column, the lower arrow points in the opposite direction.

Top row: ↑ | ←

Bottom row: ↓ | ?

A) ↑

B) →

C) ↓

D) ←

TACHS-A6-003

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Each row changes an open shape to its filled form and places a + after it.

Row 1: ○ | ● +

Row 2: □ | ?

A) □ +

B) + ■

C) ■ –

D) ■ +

TACHS-A6-004

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the third cell superimposes the first two symbols.

Row 1: | and - gives +

Row 2: / and \ gives ?

- A) |
- B) -
- C) ×
- D) +

TACHS-A6-005

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. From left to right, the filled shape moves one position downward in a three-position column, wrapping after the bottom.

Cell 1: Top ●; middle ○; bottom ○

Cell 2: Top ○; middle ●; bottom ○

Cell 3: ?

- A) Top ○; middle ○; bottom ●
- B) Top ●; middle ○; bottom ○
- C) Top ○; middle ●; bottom ○
- D) Top ●; middle ●; bottom ○

TACHS-A6-006

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular paper is folded from bottom to top along the horizontal centerline. One hole is punched near the upper-left corner of the folded rectangle, away from all edges. What appears after unfolding?

- A) One hole near the upper-left corner
- B) Two holes aligned vertically on the left side
- C) Two holes aligned horizontally across the top
- D) Four holes, one near each corner

TACHS-A6-007

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded from left to right, then a hole is punched on the folded paper's right outer edge, halfway between top and bottom. What appears after unfolding?

- A) One hole at the center
- B) Two holes on the vertical centerline
- C) Two holes, one on each outer side edge
- D) Four holes, one near each corner

TACHS-A6-008

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from top to bottom and then folded again from top to bottom. A hole is punched near the center of the folded strip, away from every edge. What appears after unfolding?

- A) Two holes in one vertical column
- B) Four holes in one vertical column
- C) Eight holes in two rows
- D) Two holes in one horizontal row

TACHS-A6-009

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded once along the diagonal from lower-left to upper-right. A small cut is made on the diagonal folded edge, away from both corners. What appears after unfolding?

- A) Four cuts at the outer corners
- B) Two cuts on the left edge
- C) A symmetric opening centered on that diagonal
- D) One cut only on the lower half

TACHS-A6-010

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. ○ inside □, with + outside
2. △ inside ◇, with + outside
3. ☆ inside ○, with + outside

Which choice follows the same rule?

- A) ■ inside □, with + outside
- B) □ inside △, with + outside

C) ◇ inside △, with × outside

D) ○ beside □, with + inside

TACHS-A6-011

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share one rule:

1. ↑ ↑ ↓

2. ← ← →

3. ↓ ↓ ↑

Which choice follows the same rule?

A) → → ←

B) → ← →

C) ↑ ↓ ↓

D) ← → ←

TACHS-A6-012

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following figures share one rule:

1. Outer □; inner ■; four + signs outside

2. Outer △; inner ▲; three + signs outside

3. Outer ◇; inner ◆; four + signs outside

Each open polygon contains its filled form and has one + for each corner. Which choice follows the rule?

A) Outer △; inner ▲; three + signs outside

B) Outer △; inner △; three + signs outside

C) Outer ◇; inner ◆; two + signs outside

D) Outer □; inner ■; three + signs outside

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R6-001 - Correct: D - Correct Answer: Her bus has stalled. - Explanation: The host messages Eli that her bus has stalled shortly before airtime.

TACHS-R6-002 - Correct: A - Correct Answer: He has become interested in what the baker just said. - Explanation: The follow-up question grows from attentive curiosity rather than from Eli's prepared list.

TACHS-R6-003 - Correct: B - Correct Answer: Listening closely can turn an uncomfortable pause into meaningful conversation. - Explanation: Eli succeeds when he listens and uses the pause to ask a genuine follow-up question.

TACHS-R6-004 - Correct: A - Correct Answer: avoiding active participation by relying on the list. - Explanation: Eli had used the prepared questions to avoid engaging directly with the conversation.

TACHS-R6-005 - Correct: B - Correct Answer: Chemical findings and human descriptions - Explanation: The passage states that an odor profile combines chemical findings with descriptions from volunteers.

TACHS-R6-006 - Correct: A - Correct Answer: To show that identifying components does not determine a complete mixture - Explanation: The distinction explains that knowing the molecules does not reveal their exact proportions in a reconstruction.

TACHS-R6-007 - Correct: A - Correct Answer: It introduces a preservation problem, explains methods and challenges, and describes possible uses. - Explanation: The passage moves from the problem of disappearing scents through collection methods, limitations, and benefits.

TACHS-R6-008 - Correct: B - Correct Answer: A wooden artifact develops an odor associated with chemical breakdown. - Explanation: Because odor changes may signal decay, a scent associated with breakdown would merit investigation.

TACHS-R6-009 - Correct: C - Correct Answer: Which passages drew attention or how an idea developed - Explanation: The third paragraph explicitly states that annotations can reveal attention and the development of ideas.

TACHS-R6-010 - Correct: B - Correct Answer: difference. - Explanation: The word refers to the difference between an earlier comment and the reader's later understanding.

TACHS-R6-011 - Correct: B - Correct Answer: Thoughtful annotations record and support a reader's changing engagement with a text. - Explanation: The passage presents annotation as evidence of active, evolving thought in print or digital form.

TACHS-R6-012 - Correct: A - Correct Answer: A reader's response provides evidence but remains one interpretation. - Explanation: The warning limits what can be concluded from a personal response to a passage.

TACHS-R6-013 - Correct: D - Correct Answer: appreciative but qualified. - Explanation: The author values useful notes while acknowledging property rules and the limits of personal interpretation.

WRITTEN EXPRESSION ANSWERS

TACHS-WE6-001 - Correct: B - Correct Answer: Sentence 4 - Explanation: Sentence 4 is a broad fact about gardening that does not develop the seed-library plan.

TACHS-WE6-002 - Correct: A - Correct Answer: Because clear labels help users choose seeds and volunteers remove old packets, volunteers will explain how to record the plant name and collection year. - Explanation: The complex sentence clearly connects the reason for labeling with the information volunteers will request.

TACHS-WE6-003 - Correct: B - Correct Answer: The packets will remain in airtight containers in a cool, dry cabinet away from sunlight. - Explanation: Choice B specifies what is stored and provides clear, practical storage conditions.

TACHS-WE6-004 - Correct: B - Correct Answer: After sentence 1 - Explanation: After sentence 1, the sentence adds an operational detail immediately following the introduction of the library.

TACHS-WE6-005 - Correct: C - Correct Answer: Working in teams, the students recorded light readings at six locations. - Explanation: The revision places "the students" directly after the modifying phrase "Working in teams."

TACHS-WE6-006 - Correct: B - Correct Answer: In addition, - Explanation: Sentence 4 adds another kind of observation, so "In addition" expresses the relationship clearly.

TACHS-WE6-007 - Correct: C - Correct Answer: The garden behind the library was dimmer than every other location. - Explanation: "Than" creates the clear and correct comparison between the garden and the other locations.

TACHS-WE6-008 - Correct: C - Correct Answer: A second evening of measurements would show whether the pattern remains consistent. - Explanation: Choice C removes repetition while preserving the proposed purpose of another measurement session.

TACHS-WE6-009 - Correct: B - Correct Answer: One of the maps shows a trail beside the river. - Explanation: The singular subject "one" correctly takes the verb "shows."

TACHS-WE6-010 - Correct: D - Correct Answer: The rain stopped before noon, and the field dried quickly. - Explanation: Choice D contains two complete clauses joined correctly by a comma and coordinating conjunction.

TACHS-WE6-011 - Correct: C - Correct Answer: Please send the completed form to him and me. - Explanation: "Him" and "me" are the correct object pronouns after the preposition "to."

TACHS-WE6-012 - Correct: D - Correct Answer: Line 1 - Explanation: Line 1 has an agreement error; the singular subject "club" requires "meets."

TACHS-WE6-013 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 requires the past-tense verb "checked" to match the passage's established tense.

MATHEMATICS ANSWERS

TACHS-M6-001 - Correct: C - Correct Answer: 15/16 - Explanation: Dividing by 14/15 means multiplying by 15/14: $\frac{7}{8} \times \frac{15}{14} = \frac{15}{16}$.

TACHS-M6-002 - Correct: D - Correct Answer: 9 - Explanation: Adding 9 gives $5x = 45$, so $x = 9$.

TACHS-M6-003 - Correct: A - Correct Answer: 22 - Explanation: The ordered list has five values, and the middle value is 22.

TACHS-M6-004 - Correct: C - Correct Answer: 44 centimeters - Explanation: Circumference is $2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ centimeters.

TACHS-M6-005 - Correct: B - Correct Answer: 3,500 milliliters - Explanation: Since 1 liter equals 1,000 milliliters, 3.5 liters equals 3,500 milliliters.

TACHS-M6-006 - Correct: D - Correct Answer: 36 - Explanation: 36 is divisible by both 12 and 18.

TACHS-M6-007 - Correct: D - Correct Answer: Multiply by 2, then add 1. - Explanation: Each term is twice the preceding term plus 1.

TACHS-M6-008 - Correct: C - Correct Answer: 1/2 - Explanation: The factors shown are 1, 2, 4, and 8, so 4 of 8 outcomes give 1/2.

TACHS-M6-009 - Correct: D - Correct Answer: 75° - Explanation: Triangle angles total 180° , and $180 - 38 - 67 = 75^\circ$.

TACHS-M6-010 - Correct: B - Correct Answer: 6 kilometers - Explanation: Multiply 1.2 kilometers per lap by 5 laps to get 6 kilometers.

TACHS-M6-011 - Correct: D - Correct Answer: \$68 - Explanation: The discount is $0.15 \times \$80 = \12 , so the sale price is \$68.

TACHS-M6-012 - Correct: C - Correct Answer: $2n - 3 \geq 15$ - Explanation: "Twice n" is $2n$, "three less" gives $2n - 3$, and "at least" means $\geq$.

ABILITY ANSWERS

TACHS-A6-001 - Correct: D - Correct Answer: $\triangle \triangle \triangle$ - Explanation: The right cell contains exactly one more open shape, so three triangles become four.

TACHS-A6-002 - Correct: B - Correct Answer: $\rightarrow$ - Explanation: The direction opposite $\leftarrow$ is $\rightarrow$.

TACHS-A6-003 - Correct: D - Correct Answer: $\blacksquare +$ - Explanation: The open square becomes filled and is followed by a plus sign.

TACHS-A6-004 - Correct: C - Correct Answer: $\times$ - Explanation: Superimposing the two diagonals / and \ forms $\times$.

TACHS-A6-005 - Correct: A - Correct Answer: Top $\bigcirc$; middle $\bigcirc$; bottom $\bullet$ - Explanation: The filled circle moves from top to middle to bottom.

TACHS-A6-006 - Correct: B - Correct Answer: Two holes aligned vertically on the left side - Explanation: The horizontal fold reflects the punch vertically, producing two left-side holes at matching heights.

TACHS-A6-007 - Correct: C - Correct Answer: Two holes, one on each outer side edge - Explanation: The punch on the outer edge reflects to the opposite outer edge after the vertical unfold.

TACHS-A6-008 - Correct: B - Correct Answer: Four holes in one vertical column - Explanation: Two horizontal folds create four vertically reflected positions when reversed.

TACHS-A6-009 - Correct: C - Correct Answer: A symmetric opening centered on that diagonal - Explanation: A cut on the fold reflects across the diagonal and forms one symmetric opening centered on it.

TACHS-A6-010 - Correct: B - Correct Answer: □ inside △, with + outside - Explanation: Each arrangement has one open shape inside a different open shape and a plus sign outside.

TACHS-A6-011 - Correct: A - Correct Answer: → → ← - Explanation: The first two arrows match, and the third points in the opposite direction.

TACHS-A6-012 - Correct: A - Correct Answer: Outer △; inner ▲; three + signs outside - Explanation: The triangle is open outside, filled inside, and accompanied by three plus signs for its three corners.

Exam Part B

Filename: tachs_full_exam_06_part_b.txt

Exam Family: TACHS

Exam Number: 6

Exam Part: B

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R6-S04

Section: Reading

Stimulus Type: Literary Passage

Title: The Borrowed Blue Thread

Text:

Minutes before the curtain rose, Nia discovered that the silver braid on the prince's costume had pulled loose. As the student responsible for wardrobe repairs, she had packed needles, pins, and thread in several colors, but no silver. The costume designer was helping an actor in another room, and the braid continued to unravel.

Nia spotted a spool of pale blue thread in the sewing box. Under the workroom lights, it clearly did not match. She carried the spool beside the stage, where the scene's blue lighting washed across the costume. Against that light, the thread nearly disappeared.

She made several small stitches instead of one long row, knotting each section separately. If one stitch broke, the others would still hold. The actor hurried onstage, and the repair survived a sword fight, a dance, and three quick costume changes.

After the show, the designer examined Nia's work. Nia began explaining why she had used the wrong color. "It was the right color for where it had to work," the designer said. She showed Nia how to replace it with silver for the next performance, but she left one blue stitch hidden beneath the braid as a reminder: a good solution depends not only on the materials available, but also on the conditions in which they will be seen.

STIMULUS END

TACHS-R6-014

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S04

Prompt:

Why does the pale blue thread appear to match during the performance?

- A) The prince's costume is entirely blue.
- B) The designer paints the thread silver.
- C) Blue stage lighting makes the thread less noticeable.
- D) The actor covers the repair with a cape.

TACHS-R6-015

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S04

Prompt:

Why does Nia knot several short sections separately?

- A) She wants the repair to remain secure if one stitch fails.
- B) She expects to remove the entire braid during the dance.
- C) She hopes the knots will reflect the stage lights.
- D) She has forgotten how to sew a continuous line.

TACHS-R6-016

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S04

Prompt:

Which statement best expresses a central idea of the passage?

- A) Temporary repairs should never be corrected later.
- B) Costume work matters more than an actor's performance.
- C) The best materials are always those that match in ordinary light.
- D) Practical solutions should account for the conditions in which they must function.

TACHS-R6-017

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S04

Prompt:

Why does the designer leave one blue stitch beneath the repaired braid?

- A) To preserve a reminder of Nia's adaptable thinking
- B) To test whether the actor notices it
- C) To avoid using any silver thread
- D) To hide a mistake from the next audience

STIMULUS START

Stimulus ID: TACHS-R6-S05

Section: Reading

Stimulus Type: Scientific Passage

Title: When Sand Sings

Text:

On a few dunes around the world, sliding sand can produce a deep hum that lasts for several seconds. The sound is sometimes called booming or singing sand. It does not occur whenever sand moves, and even nearby dunes may behave differently.

Researchers have found that grain properties matter. Singing dunes often contain dry, similarly sized grains with smooth surfaces. When a layer of these grains slides down a steep slope, many grains move in a coordinated way. Their repeated collisions create vibrations. The moving layer and the dune beneath it can reinforce particular frequencies, producing a sound much louder than separate grains would make.

Moisture can interrupt this process by changing how grains contact one another. Dust or a mixture of very different grain sizes may also prevent synchronized movement. Wind helps shape a dune and restore steep slopes, but wind alone does not create the booming sound.

Scientists still investigate how a dune's size and internal layers affect its pitch. Measurements suggest that thicker moving layers tend to produce lower sounds. By comparing dunes, researchers can test how grain motion, slope, and dune structure combine to turn countless tiny impacts into one powerful tone.

STIMULUS END

TACHS-R6-018

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S05

Prompt:

Which condition is commonly found in singing dunes?

- A) Wet grains of many different sizes
- B) Large stones mixed throughout the sand
- C) Dry grains that are similar in size
- D) Flat slopes protected from movement

TACHS-R6-019

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S05

Prompt:

As used in the second paragraph, "reinforce" most nearly means

- A) separate completely.
- B) measure precisely.
- C) change direction.
- D) make stronger.

TACHS-R6-020

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R6-S05

Prompt:

How does the third paragraph contribute to the passage?

- A) It provides a legend explaining the dunes' origin.
- B) It identifies conditions that can interfere with coordinated grain motion.
- C) It argues that wind is the only cause of the sound.
- D) It compares musical instruments with desert animals.

TACHS-R6-021

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S05

Prompt:

Which change would most likely lower the pitch of a singing dune?

- A) A thinner layer of moving sand
- B) More moisture between grains
- C) A thicker layer of moving sand
- D) A flatter slope with no sliding

STIMULUS START

Stimulus ID: TACHS-R6-S06

Section: Reading

Stimulus Type: Historical-Style Passage

Title: Signals Across the Harbor

Text:

Before radios became standard on ships, busy harbors relied on visible signals to exchange information. Flags could identify a vessel, request assistance, or announce that cargo was ready. Because a flag's meaning had to be recognized from far away, successful systems used bold patterns and limited sets of agreed-upon signals.

Weather created difficulties. Fog could hide a flag, while weak wind left it hanging unreadably. Harbors therefore combined methods. Signal stations might raise geometric shapes on a mast, ring bells, fire a harmless sound signal, or use lamps after dark. Each method worked under some conditions and failed under others.

Speed also mattered. A complicated message sent one flag at a time could delay a ship entering a narrow channel. Standard codes shortened common instructions. Instead of spelling an entire warning, one signal could tell a captain to wait while another vessel passed.

These systems required shared knowledge. Harbor workers maintained equipment, ships carried codebooks, and crews practiced recognizing signals. The flags and shapes were simple objects, but the communication network was not simple. It depended on preparation, agreement, and several backup methods working together.

STIMULUS END

TACHS-R6-022

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S06

Prompt:

Why did signal systems use bold patterns?

- A) They remained visible through every kind of weather.
- B) They were cheaper to produce than plain flags.
- C) They showed how much cargo a ship carried.
- D) They could be recognized from a distance.

TACHS-R6-023

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S06

Prompt:

As used in the second paragraph, “combined” most nearly means

- A) repaired.
- B) translated.
- C) used together.
- D) removed.

TACHS-R6-024

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S06

Prompt:

Which statement best summarizes the passage?

- A) Harbors stopped using flags because crews refused to learn codes.
- B) Early harbor communication used shared codes and multiple methods to overcome practical limits.
- C) Sound signals were more dependable than every visual method.
- D) Ships could enter narrow channels safely without receiving instructions.

TACHS-R6-025

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S06

Prompt:

The author’s main purpose is to

- A) persuade modern ships to abandon radio.
- B) explain how coordinated signaling systems addressed communication challenges.
- C) criticize crews for carrying codebooks.
- D) entertain readers with an imaginary harbor adventure.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE6-S03

Section: Written Expression

Stimulus Type: Revision Passage

Title: A Tool-Sharing Cabinet

Text:

[1] Our apartment building could reduce waste by creating a cabinet of tools that residents borrow for small repairs. [2] Many people need a drill, level, or wrench only a few times each year. [3] Purchasing separate sets for every household uses money and storage space. [4] The cabinet should stand in the locked laundry room, the manager can supervise access there. [5] Each tool would have a numbered space and a checkout card. [6] Residents should inspect borrowed items before returning them and report any damage. [7] The laundry room has six washing machines. [8] A three-month trial would show which tools are useful and whether the checkout system works.

STIMULUS END

TACHS-WE6-014

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S03

Prompt:

Which sentence is the best opening for the passage?

- A) Sentence 2
- B) Sentence 1
- C) Sentence 5
- D) Sentence 8

TACHS-WE6-015

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S03

Prompt:

Which revision corrects the error in sentence 4?

- A) The cabinet should stand in the locked laundry room the manager can supervise access there.
- B) The cabinet should stand in the locked laundry room, because, the manager can supervise access there.
- C) The cabinet should stand in the locked laundry room; the manager can supervise access there.
- D) Standing in the laundry room, access supervised by the manager.

TACHS-WE6-016

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S03

Prompt:

Which revision of sentence 6 is most concise?

- A) Residents, who are residents, should inspect the borrowed items before they return them back and report damage.
- B) Residents should inspect and report damage to tools before returning them.
- C) Before returning tools back, damage should be reported by residents after inspection.
- D) Inspection of borrowed tool items is something residents should do prior before returns.

TACHS-WE6-017

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S03

Prompt:

Which sentence should be removed because it is irrelevant to the proposal?

A) Sentence 3

B) Sentence 5

C) Sentence 7

D) Sentence 8

STIMULUS START

Stimulus ID: TACHS-WE6-S04

Section: Written Expression

Stimulus Type: Revision Passage

Title: Restoring a Footpath

Text:

[1] Last month, volunteers repaired a muddy footpath beside Pine Creek. [2] First, we cleared fallen branches and marked areas where water collected. [3] Next, shallow channels was dug to guide runoff away from the trail. [4] Volunteers then spread gravel over the lowest sections. [5] The crew worked carefully. [6] Finally, we installed two signs asking walkers to remain on the path while nearby plants recover. [7] Because the work altered how water moved, we plan to inspect the path after heavy rain. [8] Regular checks will help us correct small problems before they damage the trail again.

STIMULUS END

TACHS-WE6-018

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE6-S04

Prompt:

Which revision corrects the error in sentence 3?

A) Next, shallow channels has been dug to guide runoff away from the trail.

B) Next, a shallow channels was dug to guide runoff away from the trail.

C) Next, shallow channels is dug to guide runoff away from the trail.

D) Next, shallow channels were dug to guide runoff away from the trail.

TACHS-WE6-019

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S04

Prompt:

Which revision of sentence 5 provides the most useful detail?

A) The crew worked in a way that was careful.

- B) The volunteers were all doing various things very carefully.
- C) There was a crew, and it did some work there.
- D) Using rakes, the crew shaped the gravel into a firm, slightly raised surface.

TACHS-WE6-020

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S04

Prompt:

Which sentence best explains why the path will be inspected after heavy rain?

- A) Rain always falls at the same rate near Pine Creek.
- B) The inspection will reveal whether the new channels direct runoff as intended.
- C) The signs may become difficult for walkers to read.
- D) Volunteers prefer working during wet weather.

TACHS-WE6-021

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S04

Prompt:

Which revision most effectively combines sentences 7 and 8?

- A) The work altered water, regular checks after rain, correcting problems before damage again.
- B) Inspecting after heavy rain and because water moved differently and small problems.
- C) The trail could be damaged, and because we plan inspections, small problems.
- D) Because the work altered how water moved, inspections after heavy rain will help us correct small problems before they damage the trail again.

TACHS-WE6-022

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) The puppy wagged it's tail when the door opened.
- B) Neither answer are supported by the passage.
- C) The hikers packed their lunches before leaving.
- D) She performed the experiment more careful than I did.

TACHS-WE6-023

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence uses verb tense correctly?

- A) By the time the bell rang, we had finished the quiz.
- B) Before the guests arrived, Dad prepares the table.
- C) Last week, our class is visiting the planetarium.
- D) Yesterday, Mina walks home and found a lost glove.

TACHS-WE6-024

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

- Line 1: Each of the paintings has a wooden frame.
- Line 2: The largest painting hangs above the stairs.
- Line 3: Its bright colors attract many visitor.
- Line 4: A small label identifies the artist.

- A) Line 3
- B) Line 2
- C) Line 1
- D) Line 4

TACHS-WE6-025

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

- Line 1: The debate team practiced in the library.
- Line 2: Two students reviewed the opening argument.
- Line 3: Their coach listened and took notes.
- Line 4: Everyone spoke confident before the competition.

- A) Line 1
- B) Line 2
- C) Line 4
- D) Line 3

TACHS-WE6-026

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: My brother and me built a model bridge.

Line 2: We tested it with small metal weights.

Line 3: The center beam bent slightly under pressure.

Line 4: Our second design supported a heavier load.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M6-013

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $6 + 3 \times (8 - 5)^2$?

A) 45

B) 33

C) 81

D) 87

TACHS-M6-014

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The mean of four numbers is 18. Three numbers are 12, 17, and 25. What is the fourth number?

A) 16

B) 20

C) 18

D) 22

TACHS-M6-015

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

Two complementary angles have measures $(x + 8)^\circ$ and $(2x + 4)^\circ$. What is x ?

- A) 26
- B) 24
- C) 22
- D) 28

TACHS-M6-016

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A 10-foot board is cut into pieces measuring 2 feet 9 inches and 4 feet 5 inches. How much board remains?

- A) 2 feet 8 inches
- B) 3 feet 2 inches
- C) 2 feet 10 inches
- D) 3 feet 10 inches

TACHS-M6-017

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which fraction is equivalent to 0.375?

- A) $\frac{3}{5}$
- B) $\frac{3}{8}$
- C) $\frac{5}{8}$
- D) $\frac{7}{20}$

TACHS-M6-018

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

If $3y - 5 = 2y + 9$, what is y ?

- A) 14
- B) 9
- C) 4
- D) 18

TACHS-M6-019

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A box contains 4 pens, 3 pencils, and 5 markers. One item is selected at random. What is the probability it is not a marker?

A) $\frac{7}{12}$

B) $\frac{2}{3}$

C) $\frac{3}{4}$

D) $\frac{5}{12}$

TACHS-M6-020

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rectangular prism measures 6 cm by 4 cm by 3 cm. What is its surface area?

A) 72 cm^2

B) 108 cm^2

C) 96 cm^2

D) 84 cm^2

TACHS-M6-021

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A survey records travel times of 12, 14, 15, 15, 16, and 30 minutes. Which measure is most affected by the 30-minute value?

A) Mode

B) Median

C) Minimum

D) Mean

TACHS-M6-022

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A faucet fills 18 liters in 6 minutes at a constant rate. How many liters will it fill in 15 minutes?

- A) 36 liters
- B) 45 liters
- C) 48 liters
- D) 54 liters

TACHS-M6-023

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A number is divisible by both 6 and 15. Which must also divide the number?

- A) 10
- B) 18
- C) 20
- D) 30

TACHS-M6-024

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A taxi fare is \$4 plus \$2 for each mile. Which expression gives the fare for m miles?

- A) $2m + 4$
- B) $4m + 2$
- C) $2(m + 4)$
- D) $4(m + 2)$

TACHS-M6-025

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A square has the same perimeter as a rectangle measuring 10 units by 6 units. What is the square's area?

- A) 16 square units
- B) 36 square units
- C) 256 square units
- D) 64 square units

SECTION 4 - ABILITY

TACHS-A6-013

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the right cell reverses the order of three symbols.

Row 1: ○ △ □ | □ △ ○

Row 2: + ★ ◇ | ?

A) + ◇ ★

B) ◇ + ★

C) ★ + ◇

D) ◇ ★ +

TACHS-A6-014

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Down each column, one symbol is removed.

Top-left: ○ ○ ○

Bottom-left: ○ ○

Top-right: △ △

Bottom-right: ?

A) △ △ △

B) △

C) ▲

D) No symbol

TACHS-A6-015

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Each row alternates open, filled, open versions of one shape.

Row 1: ○ | ● | ○

Row 2: □ | ■ | ?

A) ●

B) □

C) ■

D) ○

TACHS-A6-016

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. From left to right, the line rotates 45° clockwise.

Row: | then / then ?

A) |

B) \

C) /

D) -

TACHS-A6-017

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded from right to left. Two holes are punched in the folded paper, one near the top and one near the bottom, both away from edges. What appears after unfolding?

A) Four holes in one horizontal row

B) Four holes forming two horizontal pairs

C) Two holes in one vertical line

D) Eight holes near the corners

TACHS-A6-018

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangular paper is folded from top to bottom. A triangular cut is made on the folded paper's left outer edge, away from the fold and corners. What appears after unfolding?

A) Two cuts on opposite side edges

B) One cut on the left edge

C) Two matching cuts on the left edge, mirrored vertically

D) Four cuts, one on each corner

TACHS-A6-019

Section: Ability

Type: mcq

Category: Ability
Skill: Paper Folding
Stimulus ID: None
Prompt:

A square is folded left to right, then bottom to top. A hole is punched on the folded paper's vertical outer edge, away from corners and fold lines. What appears after unfolding?

- A) Two holes on one outer edge
- B) Four holes, two on each opposite vertical edge
- C) Eight holes around all four edges
- D) Four holes at the center

TACHS-A6-020

Section: Ability
Type: mcq
Category: Ability
Skill: Paper Folding
Stimulus ID: None
Prompt:

A square is folded along the diagonal from upper-right to lower-left. Two holes are punched away from all edges at different distances from the fold. What appears after unfolding?

- A) Two holes on the diagonal
- B) Six holes around the center
- C) Four holes forming two mirrored pairs across the diagonal
- D) Three holes in one row

TACHS-A6-021

Section: Ability
Type: mcq
Category: Ability
Skill: Figure Classification
Stimulus ID: None
Prompt:

The following arrangements share one rule:

- 1. Left ○; center +; right ●
- 2. Left □; center +; right ■
- 3. Left △; center +; right ▲

Which choice follows the same rule?

- A) Left ◇; center +; right ◆
- B) Left ◆; center +; right ◇
- C) Left ☆; center ×; right ★
- D) Left ○; center +; right ■

TACHS-A6-022

Section: Ability
Type: mcq
Category: Ability
Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share one rule:

1. ○ ■ +
2. ■ + ○
3. + ○ ■

Each group contains exactly one open circle, one filled square, and one plus sign. Which choice follows the same rule?

- A) ○ □ +
- B) + ● ■
- C) ○ ■ □
- D) ■ ○ +

TACHS-A6-023

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following pairs share one rule:

1. ↑ becomes ←
2. → becomes ↑
3. ↓ becomes →

Which choice follows the same rule?

- A) ← becomes →
- B) ← becomes ↓
- C) ↑ becomes ↓
- D) → becomes ←

TACHS-A6-024

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following figures share one rule:

1. ○ outside □, with one + inside □
2. △ outside ◇, with one + inside ◇
3. ☆ outside ○, with one + inside ○

Which choice follows the same rule?

- A) ◇ outside △, with + inside △
- B) + outside □, with ◇ inside

C) ■ outside □, with + inside ■

D) □ inside △, with + outside

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R6-014 - Correct: C - Correct Answer: Blue stage lighting makes the thread less noticeable. - Explanation: The stage's blue lighting causes the pale blue thread to nearly disappear against the costume.

TACHS-R6-015 - Correct: A - Correct Answer: She wants the repair to remain secure if one stitch fails. - Explanation: Separate knots prevent one broken stitch from undoing the entire repair.

TACHS-R6-016 - Correct: D - Correct Answer: Practical solutions should account for the conditions in which they must function. - Explanation: Nia's repair succeeds because she considers stage lighting and the demands placed on the costume.

TACHS-R6-017 - Correct: A - Correct Answer: To preserve a reminder of Nia's adaptable thinking - Explanation: The hidden stitch becomes a symbol of the creative, condition-aware solution Nia found.

TACHS-R6-018 - Correct: C - Correct Answer: Dry grains that are similar in size - Explanation: The passage identifies dry, similarly sized, smooth grains as common in singing dunes.

TACHS-R6-019 - Correct: D - Correct Answer: make stronger. - Explanation: The moving layer and dune strengthen particular vibrations, making the sound louder.

TACHS-R6-020 - Correct: B - Correct Answer: It identifies conditions that can interfere with coordinated grain motion. - Explanation: The paragraph explains how moisture, dust, and varied grain sizes can disrupt synchronized movement.

TACHS-R6-021 - Correct: C - Correct Answer: A thicker layer of moving sand - Explanation: The passage states that thicker moving layers tend to produce lower sounds.

TACHS-R6-022 - Correct: D - Correct Answer: They could be recognized from a distance. - Explanation: Bold patterns made signals easier to distinguish from far away.

TACHS-R6-023 - Correct: C - Correct Answer: used together. - Explanation: Harbors used several signaling methods together because each had limitations.

TACHS-R6-024 - Correct: B - Correct Answer: Early harbor communication used shared codes and multiple methods to overcome practical limits. - Explanation: The passage emphasizes standardized codes, trained users, and backup methods for reliable communication.

TACHS-R6-025 - Correct: B - Correct Answer: explain how coordinated signaling systems addressed communication challenges. - Explanation: The passage explains the challenges, tools, codes, and cooperation involved in harbor signaling.

WRITTEN EXPRESSION ANSWERS

TACHS-WE6-014 - Correct: B - Correct Answer: Sentence 1 - Explanation: Sentence 1 introduces the proposed tool cabinet and its purpose, preparing readers for the supporting details.

TACHS-WE6-015 - Correct: C - Correct Answer: The cabinet should stand in the locked laundry room; the manager can supervise access there. - Explanation: A semicolon correctly joins the two closely related independent clauses.

TACHS-WE6-016 - Correct: B - Correct Answer: Residents should inspect and report damage to tools before returning them. - Explanation: Choice B states both responsibilities clearly and removes unnecessary repetition.

TACHS-WE6-017 - Correct: C - Correct Answer: Sentence 7 - Explanation: The number of washing machines does not help explain the tool-sharing system.

TACHS-WE6-018 - Correct: D - Correct Answer: Next, shallow channels were dug to guide runoff away from the trail. - Explanation: The plural subject "channels" requires the plural past-tense verb "were."

TACHS-WE6-019 - Correct: D - Correct Answer: Using rakes, the crew shaped the gravel into a firm, slightly raised surface. - Explanation: Choice D replaces a vague statement with a specific action that develops the repair process.

TACHS-WE6-020 - Correct: B - Correct Answer: The inspection will reveal whether the new channels direct runoff as intended. - Explanation: The added sentence directly connects later inspection to evaluating the drainage work.

TACHS-WE6-021 - Correct: D - Correct Answer: Because the work altered how water moved, inspections after heavy rain will help us correct small problems before they damage the trail again. - Explanation: Choice D clearly expresses the cause, planned inspection, and purpose in one complete sentence.

TACHS-WE6-022 - Correct: C - Correct Answer: The hikers packed their lunches before leaving. - Explanation: The plural subject and possessive pronoun agree, and the sentence uses standard grammar.

TACHS-WE6-023 - Correct: A - Correct Answer: By the time the bell rang, we had finished the quiz. - Explanation: Past perfect "had finished" correctly shows an action completed before the bell rang.

TACHS-WE6-024 - Correct: A - Correct Answer: Line 3 - Explanation: Line 3 requires the plural noun "visitors" after the quantity word "many."

TACHS-WE6-025 - Correct: C - Correct Answer: Line 4 - Explanation: Line 4 requires the adverb "confidently" to modify the verb "spoke."

TACHS-WE6-026 - Correct: A - Correct Answer: Line 1 - Explanation: Line 1 requires the subject pronoun "I": "My brother and I built."

MATHEMATICS ANSWERS

TACHS-M6-013 - Correct: B - Correct Answer: 33 - Explanation: Compute parentheses and exponent first: $6 + 3 \times 9 = 33$.

TACHS-M6-014 - Correct: C - Correct Answer: 18 - Explanation: The required total is $4 \times 18 = 72$; $72 - 54 = 18$.

TACHS-M6-015 - Correct: A - Correct Answer: 26 - Explanation: Complementary angles total 90° , so $3x + 12 = 90$ and $x = 26$.

TACHS-M6-016 - Correct: C - Correct Answer: 2 feet 10 inches - Explanation: The cuts total 7 feet 2 inches; $10 \text{ feet} - 7 \text{ feet } 2 \text{ inches} = 2 \text{ feet } 10 \text{ inches}$.

TACHS-M6-017 - Correct: B - Correct Answer: $\frac{3}{8}$ - Explanation: $0.375 = \frac{375}{1000}$, which simplifies to $\frac{3}{8}$.

TACHS-M6-018 - Correct: A - Correct Answer: 14 - Explanation: Subtracting $2y$ and adding 5 gives $y = 14$.

TACHS-M6-019 - Correct: A - Correct Answer: $\frac{7}{12}$ - Explanation: There are 7 non-markers among 12 items, so the probability is $\frac{7}{12}$.

TACHS-M6-020 - Correct: B - Correct Answer: 108 cm^2 - Explanation: Surface area is $2(6 \times 4 + 6 \times 3 + 4 \times 3) = 2(54) = 108 \text{ cm}^2$.

TACHS-M6-021 - Correct: D - Correct Answer: Mean - Explanation: The unusually high value pulls the mean upward more than it affects the median or mode.

TACHS-M6-022 - Correct: B - Correct Answer: 45 liters - Explanation: The rate is 3 liters per minute, so 15 minutes gives 45 liters.

TACHS-M6-023 - Correct: D - Correct Answer: 30 - Explanation: A number divisible by both 6 and 15 must be divisible by their least common multiple, 30.

TACHS-M6-024 - Correct: A - Correct Answer: $2m + 4$ - Explanation: The fixed \$4 charge plus \$2 per mile is represented by $4 + 2m$.

TACHS-M6-025 - Correct: D - Correct Answer: 64 square units - Explanation: The rectangle's perimeter is 32, so the square's side is 8 and its area is 64.

ABILITY ANSWERS

TACHS-A6-013 - Correct: D - Correct Answer: $\diamond \star +$ - Explanation: Reversing $+, \star, \diamond$ gives $\diamond, \star, +$.

TACHS-A6-014 - Correct: B - Correct Answer: $\triangle$ - Explanation: The lower cell contains one fewer matching open symbol, so two triangles become one triangle.

TACHS-A6-015 - Correct: B - Correct Answer: $\square$ - Explanation: The square row must finish with the open square, matching the open-filled-open pattern.

TACHS-A6-016 - Correct: D - Correct Answer: - - Explanation: After vertical | rotates 45° clockwise to /, another 45° clockwise rotation gives horizontal -.

TACHS-A6-017 - Correct: B - Correct Answer: Four holes forming two horizontal pairs - Explanation: The vertical unfold mirrors each of the two punches, producing two horizontal pairs and four holes total.

TACHS-A6-018 - Correct: C - Correct Answer: Two matching cuts on the left edge, mirrored vertically - Explanation: The horizontal unfold reflects the edge cut to a second location on the same left outer edge.

TACHS-A6-019 - Correct: B - Correct Answer: Four holes, two on each opposite vertical edge - Explanation: The horizontal fold doubles the hole vertically, and the vertical fold mirrors both to the opposite outer edge.

TACHS-A6-020 - Correct: C - Correct Answer: Four holes forming two mirrored pairs across the diagonal - Explanation: Each off-fold punch reflects once across the diagonal, producing two mirrored pairs and four holes.

TACHS-A6-021 - Correct: A - Correct Answer: Left ◇; center +; right ◆ - Explanation: An open shape appears left of + and its matching filled form appears right of +.

TACHS-A6-022 - Correct: D - Correct Answer: ■ ○ + - Explanation: Choice D contains exactly one open circle, one filled square, and one plus sign.

TACHS-A6-023 - Correct: B - Correct Answer: ← becomes ↓ - Explanation: Each arrow rotates 90° counterclockwise; ← therefore becomes ↓.

TACHS-A6-024 - Correct: A - Correct Answer: ◇ outside △, with + inside △ - Explanation: An open shape stands outside a different open shape that contains one plus sign.

Exam Part C

Filename: tachs_full_exam_06_part_c.txt

Exam Family: TACHS

Exam Number: 6

Exam Part: C

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R6-S07

Section: Reading

Stimulus Type: Literary Passage

Title: The Lantern Test

Text:

Mara's class had built paper lanterns for the courtyard festival. Her lantern was a tall hexagon covered with thin green paper. In the classroom it stood perfectly, but outside, the first breeze pushed one side inward.

Mara's first thought was to add heavier cardboard to every panel. Mr. Chen, the art teacher, asked her to carry the lantern across the courtyard before changing it. As she walked, Mara noticed that the wind struck the broadest face directly. When she turned the lantern so that one narrow corner faced the breeze, air moved around both sides and the paper stopped bending.

The display stands could not be rotated every time the wind shifted. Mara cut small openings beneath the folded decorations near the top and bottom. From a distance, the openings remained hidden, but they allowed trapped air to escape. She tested the lantern with a fan at three speeds and reinforced only the two joints that still flexed.

At the festival, the green paper trembled without collapsing. Mr. Chen asked whether the lantern was stronger. Mara considered the question. It contained very little additional material. "It is better at letting the wind pass," she said. She had begun by trying to defeat the breeze and ended by designing a path for it.

STIMULUS END

TACHS-R6-026

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S07

Prompt:

What happens when Mara points a narrow corner toward the breeze?

- A) Air moves around the lantern and the paper stops bending.
- B) The folded decorations tear away from the top.
- C) The cardboard panels become heavier.
- D) The lantern rotates back toward its broadest face.

TACHS-R6-027

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S07

Prompt:

Why does Mara test the lantern at three fan speeds?

- A) To determine which color looks brightest
- B) To dry the glue on every panel
- C) To see how the design responds to different wind strengths
- D) To make the lantern spin during the festival

TACHS-R6-028

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S07

Prompt:

Which statement best expresses a central idea of the passage?

- A) Successful designs always require more material.
- B) Outdoor artwork should be protected inside glass cases.
- C) A classroom test is more reliable than an outdoor observation.
- D) Understanding a force can lead to a solution that works with it.

TACHS-R6-029

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S07

Prompt:

As used in the third paragraph, “flexed” most nearly means

- A) changed color.
- B) came apart completely.
- C) blocked the openings.
- D) bent under pressure.

STIMULUS START

Stimulus ID: TACHS-R6-S08

Section: Reading

Stimulus Type: Literary Passage

Title: The Unfinished Pattern

Text:

Jules helped his grandfather arrange tiles for a small mosaic table. The design repeated a red diamond, two cream squares, and a blue triangle around the edge. Near the final corner, they discovered that the box contained no more blue triangles.

“We can order another,” Jules said, though the family picnic was the next morning. His grandfather spread the remaining pieces on a towel: circles, narrow rectangles, and one chipped blue square. None matched the missing tile.

Jules tried filling the space with a cream square, but the repeated pattern looked interrupted. Then he examined the chipped blue tile. He placed its broken edge against the table's outer rim and paired it with a narrow blue fragment from the scrap jar. Together, the two pieces formed the outline of a triangle, though a thin seam crossed the middle.

Jules expected his grandfather to reject the visible seam. Instead, his grandfather placed a similar seam through each of the other blue triangles using slender cream strips. The repair no longer looked like an exception; it became a feature repeated throughout the border.

At the picnic, a cousin traced the cream lines and asked where the pattern had come from. Jules glanced at the once-chipped square. "From the piece we did not have," he replied.

STIMULUS END

TACHS-R6-030

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S08

Prompt:

Which pieces does Jules combine to fill the missing space?

- A) Two cream squares
- B) A red diamond and a circle
- C) A chipped blue square and a narrow blue fragment
- D) Two complete blue triangles

TACHS-R6-031

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S08

Prompt:

Why does Jules's grandfather add cream seams to the other blue triangles?

- A) To make the repaired area part of a consistent pattern
- B) To hide every blue tile beneath cream strips
- C) To mark which tiles should be removed later
- D) To separate the border from the center of the table

TACHS-R6-032

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S08

Prompt:

The final sentence mainly emphasizes that

- A) the cousin has misunderstood the colors.
- B) the design's distinctive feature grew from a missing piece.
- C) the table should have been finished with new tiles.

D) Jules regrets using the chipped square.

TACHS-R6-033

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R6-S08

Prompt:

How does the author develop the story's resolution?

- A) By introducing a second table that has no pattern
- B) By explaining how mosaics were invented long ago
- C) By transforming a visible repair into a repeated design element
- D) By having the family postpone the picnic

STIMULUS START

Stimulus ID: TACHS-R6-S09

Section: Reading

Stimulus Type: Informational Passage

Title: Reservoirs Made of Winter

Text:

In high mountain regions, winter water may be plentiful while spring planting water is scarce. Snow and ice often melt too late or flow away before fields need irrigation. Some communities address this timing problem by building cone-shaped towers of ice during cold months.

Water from a stream or uphill pipe travels downward through a narrow pipe. The difference in elevation creates pressure, so no electric pump is required. A nozzle sprays the water into freezing air. Droplets freeze as they fall and gradually build a tall cone around a central structure.

The cone shape is useful because it exposes less surface area to sunlight than a shallow sheet holding the same amount of ice. As temperatures rise, the tower melts slowly. Channels carry the released water toward young trees or fields. Builders may adjust a tower's location and size according to shade, wind, expected temperatures, and local water needs.

These ice towers do not increase the total amount of water in a watershed. Instead, they temporarily store part of a winter flow and release it later. Their success depends on sufficiently cold construction weather and careful placement. Where conditions fit, the towers serve as seasonal reservoirs built from the water they are designed to save.

STIMULUS END

TACHS-R6-034

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S09

Prompt:

What creates pressure that moves water through the pipe?

- A) Heat absorbed by the central structure
- B) An electric pump inside the tower
- C) A difference in elevation
- D) Wind striking the nozzle

TACHS-R6-035

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S09

Prompt:

As used in the final paragraph, “fit” most nearly means

- A) be measured exactly.
- B) increase in size.
- C) become warmer.
- D) are suitable.

TACHS-R6-036

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S09

Prompt:

Which statement best summarizes the passage?

- A) Cone-shaped ice towers delay part of winter water flow for later seasonal use.
- B) Electric pumps carry spring water into winter storage tanks.
- C) Ice towers create new water by freezing mountain air.
- D) Shallow sheets of ice always melt more slowly than towers.

TACHS-R6-037

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S09

Prompt:

Which site would most likely allow an ice tower to last longer into spring?

- A) An open, sunny slope with warm winds
- B) A low field where winter temperatures remain above freezing
- C) A dark indoor room far from any water supply
- D) A shaded location protected from strong warm winds

TACHS-R6-038

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S09

Prompt:

Why does the author state that the towers do not increase total watershed water?

- A) To clarify that the method changes the timing of water availability, not its amount

- B) To argue that the towers have no useful purpose
- C) To show that the ice is made from rain outside the region
- D) To suggest that every winter stream should be blocked

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE6-S05

Section: Written Expression

Stimulus Type: Revision Passage

Title: Recording an Audio Tour

Text:

[1] Our history club is creating a short audio tour of the old train depot. [2] Visitors will scan a code at five marked locations to hear a one-minute recording. [3] Each recording should identify one visible feature, explain why it mattered, and include a question that encourages visitors to look closely. [4] Long lists of dates are available in many reference books. [5] To keep the narration natural, students will practice reading aloud before recording. [6] Background sounds will be recorded separately, they can be adjusted without covering the speaker's voice. [7] Finally, another class will test the tour and report any confusing directions. [8] Their feedback will help us revise the recordings before the public opening.

STIMULUS END

TACHS-WE6-027

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S05

Prompt:

Which sentence should be removed because it does not develop the audio-tour plan?

- A) Sentence 2
- B) Sentence 7
- C) Sentence 5
- D) Sentence 4

TACHS-WE6-028

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S05

Prompt:

Which revision corrects the error in sentence 6?

- A) Background sounds will be recorded separately, and they can be adjusted without covering the speaker's voice.
- B) Background sounds recorded separately, adjusting without covering the speaker's voice.
- C) Because background sounds will be recorded separately and can be adjusted.
- D) Background sounds will be recorded separately they can be adjusted, without covering the voice.

TACHS-WE6-029

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S05

Prompt:

Which revision of sentence 3 is most concise while preserving all key ideas?

- A) Each and every recording should identify one feature that is visible to the eye and explain why it was important in importance.
- B) One visible feature should be identified in each recording, and an explanation and looking question should also be included.
- C) The recordings should have visible things, explanations, and a question of some kind for people.
- D) Each recording should point out a visible feature, explain its importance, and ask listeners to observe it closely.

TACHS-WE6-030

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S05

Prompt:

Which sentence would best follow sentence 2?

- A) The depot roof was repaired twelve years ago.
- B) Club meetings usually end at four o'clock.
- C) Some visitors prefer headphones with soft ear cushions.
- D) A map at the entrance will show the numbered route between those locations.

STIMULUS START

Stimulus ID: TACHS-WE6-S06

Section: Written Expression

Stimulus Type: Revision Passage

Title: Signs for More Than One Festival

Text:

[1] The neighborhood arts festival needs directional signs every year. [2] In the past, organizers printed dates directly on the signs, so most could not be reused. [3] This year, volunteers painted permanent arrows and place names on sturdy boards. [4] Dates and times appear on removable panels attached with clips. [5] After the festival, the boards are stored flat in labeled sleeves. [6] Next year, only the small information panels will need to be replaced. [7] Reusing the main signs reduces both material waste and the hours spending on painting. [8] The new system also allows a damaged panel to be replaced without discarding an entire sign.

STIMULUS END

TACHS-WE6-031

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE6-S06

Prompt:

Which revision corrects the error in sentence 7?

- A) Reusing the main signs reduces both material waste and the hours spend on painting.
- B) Reusing the main signs reduce both material waste and the hours spent on painting.
- C) Reusing the main signs reduces both material waste and the hours spent on painting.
- D) Reusing the main signs reduces both material waste and the hours spending on painting.

TACHS-WE6-032

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S06

Prompt:

Which revision most effectively combines sentences 5 and 6?

- A) Stored flat in labeled sleeves after the festival, the boards can be reused next year with only the small information panels replaced.
- B) The boards are in sleeves because next year and replaced panels.
- C) Storing boards, the information panels will replace them next year.
- D) After the festival, the boards stored flat, and next year small panels only.

TACHS-WE6-033

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S06

Prompt:

Which phrase best replaces “The new system” in sentence 8 to make the reference more precise?

- A) The whole idea mentioned
- B) That sort of thing
- C) This modular sign design
- D) It

TACHS-WE6-034

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) The blue backpack is heavier than the red one.
- B) There are less chairs in this room than in the auditorium.
- C) Of the two routes, this one is the shortest.
- D) Jamal completed the puzzle more quicker than I did.

TACHS-WE6-035

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: None

Prompt:

Which sentence uses parallel structure correctly?

- A) The committee planned the event, assigning the tasks, and the schedule was confirmed.
- B) The committee planned the event, assigned the tasks, and confirmed the schedule.
- C) The committee was planning the event, to assign tasks, and confirmed the schedule.
- D) Planning the event, the tasks were assigned, and schedule confirmation.

TACHS-WE6-036

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: None

Prompt:

Which sentence is most appropriate for a formal request to a principal?

- A) Hey, can you give us the gym Friday?
- B) Our club respectfully requests permission to use the gym on Friday afternoon.
- C) The gym would be cool for our thing on Friday.
- D) We gotta use the gym, so let us know.

TACHS-WE6-037

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The package arrived earlier than expected.

Line 2: Its contents were protected by thick paper.

Line 3: We opened the box careful with a small knife.

Line 4: Nothing inside had been damaged.

- A) Line 3
- B) Line 2
- C) Line 1
- D) Line 4

TACHS-WE6-038

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: Before sunrise, the campers packed their tents.

Line 2: Luis checked that no equipment had been left behind.

Line 3: The group followed the marked trail toward the lake.

Line 4: At noon, they will reach the ranger station.

A) Line 1

B) Line 2

C) Line 4

D) Line 3

SECTION 3 - MATHEMATICS

TACHS-M6-026

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is 35% of 240?

A) 72

B) 96

C) 84

D) 105

TACHS-M6-027

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

If $2a + 3b = 19$ and $a = 5$, what is b ?

A) 3

B) 2

C) 4

D) 5

TACHS-M6-028

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The temperatures were 14° , 18° , 18° , 21° , 24° , and 25° . What is the range?

A) 7°

- B) 9°
- C) 39°
- D) 11°

TACHS-M6-029

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A parallelogram has base 13 cm and height 6 cm. What is its area?

- A) 78 cm^2
- B) 39 cm^2
- C) 72 cm^2
- D) 38 cm^2

TACHS-M6-030

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A movie begins at 6:38 p.m. and lasts 1 hour 47 minutes. When does it end?

- A) 9:25 p.m.
- B) 8:15 p.m.
- C) 8:35 p.m.
- D) 8:25 p.m.

TACHS-M6-031

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which value is closest to $\sqrt{70}$?

- A) 7.1
- B) 9.2
- C) 8.4
- D) 10.1

TACHS-M6-032

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $3(x + 4) - 2 = 25$.

- A) 3
- B) 4
- C) 7
- D) 5

TACHS-M6-033

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A jar has 5 white and 7 black beads. A white bead is drawn and not replaced. What is the probability the next bead is black?

- A) $\frac{7}{11}$
- B) $\frac{5}{12}$
- C) $\frac{7}{12}$
- D) $\frac{5}{11}$

TACHS-M6-034

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A square's diagonal divides it into two congruent triangles. If the square's area is 98 square units, what is the area of one triangle?

- A) 24.5 square units
- B) 49 square units
- C) 56 square units
- D) 196 square units

TACHS-M6-035

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A scale drawing uses 2 centimeters for 5 meters. A wall measures 7 centimeters in the drawing. What is its actual length?

- A) 14 meters
- B) 17.5 meters

C) 15 meters

D) 35 meters

TACHS-M6-036

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the least prime number greater than 40?

A) 45

B) 43

C) 41

D) 47

TACHS-M6-037

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A function follows the rule $f(n) = n^2 - 2$. What is $f(6)$?

A) 32

B) 36

C) 34

D) 38

SECTION 4 - ABILITY

TACHS-A6-025

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the right cell contains the two symbols from the left cell with their positions exchanged.

Row 1 left: Top ○; bottom ■

Row 1 right: Top ■; bottom ○

Row 2 left: Top △; bottom ◆

Row 2 right: ?

A) Top ◆; bottom △

B) Top △; bottom ◆

C) Top ▲; bottom ◇

D) Top ◇; bottom ▲

TACHS-A6-026

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the number of shapes increases by two.

Row 1: ○ | ○ ○ ○

Row 2: □ □ | ?

A) □ □ □

B) □ □ □ □ □

C) □ □ □ □

D) ■ ■ ■ ■

TACHS-A6-027

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Down each column, a + changes to × while the accompanying shape remains unchanged.

Top-left: ○ +

Bottom-left: ○ ×

Top-right: □ +

Bottom-right: ?

A) ■ ×

B) □ ×

C) □ +

D) ■ +

TACHS-A6-028

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the arrow turns 180° and the open shape becomes filled.

Row 1: ↑ ○ | ↓ ●

Row 2: ← □ | ?

A) ← ■

B) → ■

C) → □

D) ↑ ■

TACHS-A6-029

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the third cell contains the symbol common to the first two cells.

Row 1: {○, □} | {□, △} | □

Row 2: {+, ★} | {★, ×} | ?

A) ☆

B) +

C) ×

D) ★

TACHS-A6-030

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from left to right. A hole is punched exactly on the horizontal centerline of the folded rectangle, away from edges and the vertical fold. What appears after unfolding?

A) One hole on the vertical fold

B) Four holes around the center

C) Two holes in a vertical column

D) Two holes in a horizontal row, symmetric across the vertical centerline

TACHS-A6-031

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangle is folded from right to left, then from top to bottom. A hole is punched near the lower-left corner of the folded rectangle, away from all edges. What appears after unfolding?

A) Two holes on the left side

B) Four holes on the horizontal centerline

C) Four holes, one in each corresponding quadrant

D) Eight holes near the perimeter

TACHS-A6-032

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded once from bottom to top. A semicircular cut is made on the folded paper's top outer edge, away from corners. What appears after unfolding?

- A) One complete circle at the horizontal centerline
- B) Two complete circles in a vertical column
- C) Two matching semicircular cuts, one on the top edge and one on the bottom edge
- D) One semicircle on the bottom edge

TACHS-A6-033

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from left to right. The resulting rectangle is then folded along its diagonal from lower-left to upper-right, bringing its upper-left corner onto its lower-right corner. One hole is punched away from all edges and fold lines. What appears after fully unfolding?

- A) Eight holes along the edges
- B) Two holes on the original diagonal
- C) Six holes around one corner
- D) Four holes related by the two fold reflections

TACHS-A6-034

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. ○ above □, with a vertical line between them
2. △ above ◇, with a vertical line between them
3. ☆ above ○, with a vertical line between them

Which choice follows the same rule?

- A) □ above △, with a vertical line between
- B) ◇ above △, with a horizontal line between
- C) ■ above □, with a vertical line between
- D) ○ beside □, with a vertical line between

TACHS-A6-035

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following sequences share one rule:

1. ○ □ △
2. □ △ ○
3. △ ○ □

Which choice follows the same rule?

- A) ○ □ △
- B) □ ○ △
- C) △ □ ○
- D) ○ △ □

TACHS-A6-036

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following figures share one rule:

1. Outer ○; two identical + signs inside
2. Outer □; two identical × signs inside
3. Outer △; two identical ★ signs inside

Which choice follows the same rule?

- A) Outer ◆; two identical ○ signs inside
- B) Outer ◇; two identical ○ signs inside
- C) Outer ◇; ○ and ● inside
- D) Outer ◇; one ○ inside

TACHS-A6-037

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. Left ●; right ○
2. Left ■; right □
3. Left ▲; right △

Which choice follows the same rule?

- A) Left ◇; right ◆
- B) Left ○; right ●
- C) Left ★; right ☆
- D) Left ◆; right □

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R6-026 - Correct: A - Correct Answer: Air moves around the lantern and the paper stops bending. - Explanation: Pointing a corner into the breeze allows air to flow around both sides and reduces pressure on a panel.

TACHS-R6-027 - Correct: C - Correct Answer: To see how the design responds to different wind strengths - Explanation: Testing several speeds helps Mara evaluate the revised design under varied wind conditions.

TACHS-R6-028 - Correct: D - Correct Answer: Understanding a force can lead to a solution that works with it. - Explanation: Mara improves the lantern by redirecting airflow rather than simply resisting the wind with added weight.

TACHS-R6-029 - Correct: D - Correct Answer: bent under pressure. - Explanation: The joints that flexed bent when exposed to moving air.

TACHS-R6-030 - Correct: C - Correct Answer: A chipped blue square and a narrow blue fragment - Explanation: Jules pairs the chipped blue square with a narrow blue fragment to create a triangular outline.

TACHS-R6-031 - Correct: A - Correct Answer: To make the repaired area part of a consistent pattern - Explanation: Repeating the seam turns a necessary repair into an intentional feature of the whole design.

TACHS-R6-032 - Correct: B - Correct Answer: the design's distinctive feature grew from a missing piece. - Explanation: Jules recognizes that the shortage inspired the repeated seam that now defines the pattern.

TACHS-R6-033 - Correct: C - Correct Answer: By transforming a visible repair into a repeated design element - Explanation: The grandfather repeats the repair's seam, resolving the mismatch by incorporating it into the design.

TACHS-R6-034 - Correct: C - Correct Answer: A difference in elevation - Explanation: Water moving from a higher elevation creates pressure without an electric pump.

TACHS-R6-035 - Correct: D - Correct Answer: are suitable. - Explanation: The phrase refers to locations where weather and placement conditions are suitable for the towers.

TACHS-R6-036 - Correct: A - Correct Answer: Cone-shaped ice towers delay part of winter water flow for later seasonal use. - Explanation: The passage explains how conical ice structures store winter flow and release it when water is needed later.

TACHS-R6-037 - Correct: D - Correct Answer: A shaded location protected from strong warm winds - Explanation: Shade and protection from warm wind would slow melting, assuming winter construction conditions are cold enough.

TACHS-R6-038 - Correct: A - Correct Answer: To clarify that the method changes the timing of water availability, not its amount - Explanation: The statement prevents a misconception by distinguishing seasonal storage from creation of additional water.

WRITTEN EXPRESSION ANSWERS

TACHS-WE6-027 - Correct: D - Correct Answer: Sentence 4 - Explanation: Sentence 4 mentions reference books but does not explain the tour's content, production, or testing.

TACHS-WE6-028 - Correct: A - Correct Answer: Background sounds will be recorded separately, and they can be adjusted without covering the speaker's voice. - Explanation: A comma and coordinating conjunction correctly join the two independent clauses.

TACHS-WE6-029 - Correct: D - Correct Answer: Each recording should point out a visible feature, explain its importance, and ask listeners to observe it closely. - Explanation: Choice D preserves the three content goals

with clear parallel verbs and no unnecessary wording.

TACHS-WE6-030 - Correct: D - Correct Answer: A map at the entrance will show the numbered route between those locations. - Explanation: A route map logically explains how visitors will move among the five marked listening locations.

TACHS-WE6-031 - Correct: C - Correct Answer: Reusing the main signs reduces both material waste and the hours spent on painting. - Explanation: The singular gerund subject takes “reduces,” and “hours spent” uses the correct participle.

TACHS-WE6-032 - Correct: A - Correct Answer: Stored flat in labeled sleeves after the festival, the boards can be reused next year with only the small information panels replaced. - Explanation: Choice A clearly connects proper storage with reuse and avoids a misplaced modifier.

TACHS-WE6-033 - Correct: C - Correct Answer: This modular sign design - Explanation: “This modular sign design” names the system precisely and clarifies why individual panels can be replaced.

TACHS-WE6-034 - Correct: A - Correct Answer: The blue backpack is heavier than the red one. - Explanation: “Heavier than” makes a correct comparison between two backpacks.

TACHS-WE6-035 - Correct: B - Correct Answer: The committee planned the event, assigned the tasks, and confirmed the schedule. - Explanation: Choice B uses three parallel past-tense verbs: “planned,” “assigned,” and “confirmed.”

TACHS-WE6-036 - Correct: B - Correct Answer: Our club respectfully requests permission to use the gym on Friday afternoon. - Explanation: Choice B uses a specific, courteous, and appropriately formal tone.

TACHS-WE6-037 - Correct: A - Correct Answer: Line 3 - Explanation: Line 3 requires the adverb “carefully” to modify “opened.”

TACHS-WE6-038 - Correct: C - Correct Answer: Line 4 - Explanation: Line 4 incorrectly shifts to future tense; “reached” should match the past-tense context.

MATHEMATICS ANSWERS

TACHS-M6-026 - Correct: C - Correct Answer: 84 - Explanation: $0.35 \times 240 = 84$.

TACHS-M6-027 - Correct: A - Correct Answer: 3 - Explanation: Substitute 5: $10 + 3b = 19$, so $3b = 9$ and $b = 3$.

TACHS-M6-028 - Correct: D - Correct Answer: 11° - Explanation: The range is $25 - 14 = 11$ degrees.

TACHS-M6-029 - Correct: A - Correct Answer: 78 cm^2 - Explanation: Area equals base $\times$ height: $13 \times 6 = 78 \text{ cm}^2$.

TACHS-M6-030 - Correct: D - Correct Answer: 8:25 p.m. - Explanation: Adding 1 hour gives 7:38, and adding 47 minutes gives 8:25 p.m.

TACHS-M6-031 - Correct: C - Correct Answer: 8.4 - Explanation: Since $8^2 = 64$ and $9^2 = 81$, $\sqrt{70}$ is about 8.4.

TACHS-M6-032 - Correct: D - Correct Answer: 5 - Explanation: $3x + 12 - 2 = 25$, so $3x = 15$ and $x = 5$.

TACHS-M6-033 - Correct: A - Correct Answer: $7/11$ - Explanation: After a white bead is removed, 7 of the 11 remaining beads are black.

TACHS-M6-034 - Correct: B - Correct Answer: 49 square units - Explanation: A diagonal divides the square into equal halves, so each triangle has area $98 \div 2 = 49$.

TACHS-M6-035 - Correct: B - Correct Answer: 17.5 meters - Explanation: Each centimeter represents 2.5 meters, so $7 \times 2.5 = 17.5$ meters.

TACHS-M6-036 - Correct: C - Correct Answer: 41 - Explanation: 41 has no positive factors other than 1 and 41 and is the first integer above 40.

TACHS-M6-037 - Correct: C - Correct Answer: 34 - Explanation: $f(6) = 6^2 - 2 = 36 - 2 = 34$.

ABILITY ANSWERS

TACHS-A6-025 - Correct: A - Correct Answer: Top $\blacklozenge$; bottom $\triangle$ - Explanation: Exchanging the top and bottom positions places $\blacklozenge$ above $\triangle$.

TACHS-A6-026 - Correct: C - Correct Answer: $\square \square \square \square$ - Explanation: Increasing two squares by two gives four open squares.

TACHS-A6-027 - Correct: B - Correct Answer: $\square \times$ - Explanation: The square stays open while $+$ changes to $\times$.

TACHS-A6-028 - Correct: B - Correct Answer: → ■ - Explanation: A 180° turn changes ← to →, and open □ becomes filled ■.

TACHS-A6-029 - Correct: D - Correct Answer: ★ - Explanation: The only symbol shared by the two Row 2 sets is ★.

TACHS-A6-030 - Correct: D - Correct Answer: Two holes in a horizontal row, symmetric across the vertical centerline - Explanation: The vertical unfold mirrors the off-fold punch horizontally; its centerline height remains unchanged.

TACHS-A6-031 - Correct: C - Correct Answer: Four holes, one in each corresponding quadrant - Explanation: Two perpendicular folds reflect the punch across both centerlines, producing four corresponding positions.

TACHS-A6-032 - Correct: C - Correct Answer: Two matching semicircular cuts, one on the top edge and one on the bottom edge - Explanation: The outer-edge cut reflects across the horizontal centerline to the opposite outer edge.

TACHS-A6-033 - Correct: D - Correct Answer: Four holes related by the two fold reflections - Explanation: Each of the two folds doubles the off-edge punch, yielding four reflected positions.

TACHS-A6-034 - Correct: A - Correct Answer: □ above △, with a vertical line between - Explanation: Two different open shapes are arranged vertically with a vertical line between them.

TACHS-A6-035 - Correct: A - Correct Answer: ○ □ △ - Explanation: Each listed sequence follows the same cyclic order ○, □, △; choice A is one of those valid rotations.

TACHS-A6-036 - Correct: B - Correct Answer: Outer ◇; two identical ○ signs inside - Explanation: An open outer shape contains exactly two identical symbols.

TACHS-A6-037 - Correct: C - Correct Answer: Left ★; right ☆ - Explanation: The filled form appears on the left and its matching open form appears on the right.

Exam Part D

Filename: tachs_full_exam_06_part_d.txt

Exam Family: TACHS

Exam Number: 6

Exam Part: D

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R6-S10

Section: Reading

Stimulus Type: Literary Passage

Title: The Impossible Temperature

Text:

For the science club's greenhouse project, Dev recorded the temperature every afternoon. The chart rose gently on sunny days and dipped after rain. Then Tuesday's entry showed 119°F, even though the greenhouse had felt only pleasantly warm.

Dev's partner, Ana, wanted to erase the number. "It is obviously wrong," she said. Dev agreed that the greenhouse had not reached 119°F, but he drew a box around the entry instead. They checked the digital sensor and found its screen working normally.

The next afternoon, Dev watched the sensor at the usual recording time. A narrow beam of sunlight moved across the wall and struck the black plastic case. Within minutes, the displayed temperature climbed far above the reading from a shaded thermometer nearby. The sensor was measuring heat around its sun-warmed case rather than the greenhouse air.

Dev and Ana built a small white cover that shaded the device while allowing air to circulate. They kept Tuesday's number on the chart and added a note explaining the problem. At the project fair, the unusual point attracted more questions than the smooth parts of the graph. Dev realized that a mistaken measurement could still be accurate evidence—just not evidence of what he had first assumed.

STIMULUS END

TACHS-R6-039

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S10

Prompt:

What causes the sensor's unusually high reading?

- A) A sunbeam heats its black plastic case.
- B) The greenhouse heater turns on automatically.
- C) Ana places it beside a shaded thermometer.
- D) Rainwater enters its digital screen.

TACHS-R6-040

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S10

Prompt:

Why do Dev and Ana keep Tuesday's reading on the chart?

- A) They still believe the greenhouse reached 119°F.
- B) It is the only temperature they recorded that week.
- C) They need the value to make every graph line straight.
- D) It helps document the measurement problem they discovered.

TACHS-R6-041

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S10

Prompt:

The author includes visitors' questions at the fair mainly to show that

- A) scientific displays should contain as many errors as possible.
- B) the shaded thermometer was unnecessary.
- C) investigating an unexpected result can deepen a project.
- D) visitors cared only about the highest number.

TACHS-R6-042

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S10

Prompt:

As used in the final sentence, "assumed" most nearly means

- A) measured with a second instrument.
- B) proved through repeated trials.
- C) accepted as true without first verifying.
- D) reported to the project judges.

STIMULUS START

Stimulus ID: TACHS-R6-S11

Section: Reading

Stimulus Type: Scientific Passage

Title: The Flexible Armor of a Pangolin

Text:

A pangolin's body is covered with overlapping scales made largely of keratin, the same type of protein found in human fingernails. The scales provide protection, but an animal covered by rigid plates must still be able to move.

The solution lies partly in arrangement. Each scale overlaps nearby scales like a roof tile, yet its base attaches to flexible skin. When the pangolin walks or climbs, the scales shift slightly over one another. When threatened, the animal curls into a ball. The scales spread at their exposed edges while continuing to cover the softer skin beneath them.

Shape also contributes to strength. A scale is not a flat sheet; it curves and contains layers of fibers oriented in different directions. A crack that begins in one layer may be redirected or slowed when it reaches another. This layered structure helps prevent a small defect from traveling straight through the scale.

Researchers study such biological structures for design ideas. Overlapping protective panels might be useful where equipment must resist impact without becoming completely stiff. Copying a scale exactly is not always necessary. Engineers may instead apply the broader principles: movable overlap, curved surfaces, and layered fibers that guide damage away from a direct path.

STIMULUS END

TACHS-R6-043

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S11

Prompt:

What allows neighboring scales to shift as the pangolin moves?

- A) Each scale dissolves and reforms.
- B) The scales are made of soft muscle.
- C) Air pockets inflate beneath every scale.
- D) Their bases attach to flexible skin.

TACHS-R6-044

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S11

Prompt:

Which statement best summarizes the passage?

- A) A pangolin's scales stop all cracks before they begin.
- B) Keratin is useful only when formed into flat, rigid sheets.
- C) Engineers can improve armor only by copying pangolin scales exactly.
- D) Pangolin scales combine overlap, curvature, and layers to provide protection while permitting movement.

TACHS-R6-045

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R6-S11

Prompt:

As used in the third paragraph, "redirected" most nearly means

- A) sent along a different path.
- B) hidden from view.
- C) made wider.

D) measured at a later time.

TACHS-R6-046

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R6-S11

Prompt:

Why does the author say that copying a scale exactly is not always necessary?

- A) To show that scales have no useful engineering features
- B) To emphasize that designers can adapt underlying structural principles.
- C) To prove that pangolins cannot move while curled
- D) To argue that artificial materials should never use layers

STIMULUS START

Stimulus ID: TACHS-R6-S12

Section: Reading

Stimulus Type: Biographical-Style Passage

Title: Writing Movement on Paper

Text:

Choreographer Elena Maris often created dance sequences by demonstrating them in a studio. Dancers learned by watching and repeating. When a performer asked how an older work had originally ended, however, Maris discovered that memory and a grainy video offered different answers.

She began developing a personal notation system. An arrow showed direction, a short line showed duration, and symbols above or below a horizontal mark indicated whether a movement was high or low. At first, Maris tried to record every turn of a wrist. Her pages became so crowded that even she could not read them quickly.

Maris revised the system to capture decisions rather than every detail. She recorded pathways, timing, levels, and moments when dancers could choose among several gestures. Performers still interpreted the quality of a motion, but they could reconstruct the work's structure without copying one video performance.

The notation also changed how Maris composed. By placing symbols on cards and rearranging them, she noticed repeated pathways and tested new sequences before calling dancers into the studio. Paper did not replace bodies in motion. It gave Maris another way to examine, preserve, and revise what those bodies could do.

STIMULUS END

TACHS-R6-047

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R6-S12

Prompt:

What did arrows represent in Maris's notation?

- A) Direction
- B) Duration
- C) Height

D) A dancer's name

TACHS-R6-048

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R6-S12

Prompt:

Why does Maris stop recording every small movement?

- A) The dancers refuse to use any written symbols.
- B) Excessive detail makes the notation difficult to use.
- C) Her video provides a perfect record of every performance.
- D) She decides timing is unimportant to dance.

TACHS-R6-049

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R6-S12

Prompt:

What is the central idea of the passage?

- A) Written notation makes live dancers unnecessary.
- B) A video always preserves a dance more accurately than memory.
- C) A selective notation system helps Maris preserve and develop choreography.
- D) Dance movements can be recorded only through detailed drawings.

TACHS-R6-050

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R6-S12

Prompt:

How is the passage mainly organized?

- A) A finished notation system is compared with several unrelated alphabets.
- B) A problem leads to an initial solution, revision, and additional use.
- C) Several dancers argue about who created a movement.
- D) A dance is described from opening step to final bow.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE6-S07

Section: Written Expression

Stimulus Type: Revision Passage

Title: A Repair Log for Classroom Equipment

Text:

[1] Our technology classroom contains cameras, tripods, microphones, and charging cables used by many groups. [2] When an item stops working, students sometimes place it back on the shelf without describing the problem. [3] The next group may waste time testing the same faulty equipment. [4] A repair log beside the storage cabinet would allow users to record the item number, the date, and a brief description of the problem. [5] Entries should be specific, “Microphone 4 produces a crackling sound” is more useful than “broken.” [6] The teacher could mark whether an item is awaiting inspection, being repaired, or ready to use. [7] Clear records would prevent repeated testing and help the class decide which replacements are most urgent.

STIMULUS END

TACHS-WE6-039

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S07

Prompt:

Which sentence best states the problem that the proposed log would solve?

- A) Sentence 4
- B) Sentence 1
- C) Sentence 2
- D) Sentence 6

TACHS-WE6-040

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S07

Prompt:

Which revision corrects the error in sentence 5?

- A) Entries should be specific; “Microphone 4 produces a crackling sound” is more useful than “broken.”
- B) Entries should be specific “Microphone 4 produces a crackling sound,” being more useful than “broken.”
- C) Entries should be specific, and because “Microphone 4 produces a crackling sound” is useful.
- D) Specific entries, “Microphone 4 produces a crackling sound” more useful than “broken.”

TACHS-WE6-041

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S07

Prompt:

Which phrase best replaces “stops working” in sentence 2 to make the meaning more precise?

- A) does a thing incorrectly
- B) does not function as expected
- C) is whatever is wrong

D) gets kind of weird

TACHS-WE6-042

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE6-S07

Prompt:

Which sentence would provide the best conclusion after sentence 7?

- A) The cabinet has three shelves and two doors.
- B) Cameras were invented before digital microphones.
- C) Some classes meet in the morning and others meet after lunch.
- D) Students could review the log each month to identify recurring equipment problems.

STIMULUS START

Stimulus ID: TACHS-WE6-S08

Section: Written Expression

Stimulus Type: Revision Passage

Title: Comparing Two Soil Covers

Text:

[1] We compared how straw and wood chips affected moisture in two equal garden plots. [2] Before adding the covers, we gave each plot the same amount of water. [3] One plot received a five-centimeter layer of straw; the other received an equally deep layer of wood chips. [4] Moisture was measured at the same depth each morning for ten days. [5] The straw-covered plot lost water fastest during the first three days. [6] By day ten, however, both covered plots was moister than an uncovered comparison area. [7] The trial suggests that either cover can slow moisture loss, but a longer study would be needed to compare their performance in different weather.

STIMULUS END

TACHS-WE6-043

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE6-S08

Prompt:

Which revision corrects the error in sentence 6?

- A) By day ten, however, both covered plots is moister than an uncovered comparison area.
- B) By day ten, however, both covered plots has been moister than an uncovered comparison area.
- C) By day ten, however, both covered plot was moister than an uncovered comparison area.
- D) By day ten, however, both covered plots were moister than an uncovered comparison area.

TACHS-WE6-044

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE6-S08

Prompt:

Which revision most effectively combines sentences 2 and 3?

- A) After receiving equal amounts of water, one plot was covered with five centimeters of straw and the other with five centimeters of wood chips.
- B) Before covers, water was given, one plot straw and one wood chips equally deep.
- C) The plots had water because straw, while wood chips were equal in another.
- D) We gave each plot water, the covers were straw and wood chips, they were five centimeters.

TACHS-WE6-045

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE6-S08

Prompt:

Which revision of sentence 5 is most precise?

- A) The straw plot did a different thing at first.
- B) During days 1 through 3, moisture readings decreased more quickly in the straw-covered plot.
- C) Water was lost by something during some days.
- D) The straw was faster, and that was how the plot went.

TACHS-WE6-046

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) Each of the volunteers has received a name tag.
- B) Neither of the bicycles need a new chain.
- C) The box of old photographs were stored upstairs.
- D) One of the musicians have arrived early.

TACHS-WE6-047

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) If I was taller, I would reach the shelf.
- B) She asked whether the package had arrived.
- C) The dog laid on the rug all afternoon.
- D) We should of checked the schedule earlier.

TACHS-WE6-048

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The guide described the cave's limestone walls.

Line 2: We walked slowly across the damp floor.

Line 3: Everyone kept there flashlight pointed downward.

Line 4: At the exit, we cleaned mud from our shoes.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

TACHS-WE6-049

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The recipe calls for two ripe pears.

Line 2: First, slice the fruit into thin pieces.

Line 3: Next, arrange the slices evenly in the pan.

Line 4: Bake them until the crust looked golden.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

TACHS-WE6-050

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The librarian placed the returned books on a cart.

Line 2: A volunteer sorted them by call number.

Line 3: Afterward, the shelves were filled with book's.

Line 4: The empty cart was returned to the desk.

A) Line 3

- B) Line 2
- C) Line 1
- D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M6-038

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $2\frac{2}{3} - 1\frac{5}{6}$?

- A) $\frac{1}{6}$
- B) $\frac{5}{6}$
- C) $1\frac{1}{6}$
- D) $1\frac{5}{6}$

TACHS-M6-039

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The scores are 72, 78, 81, 85, and 94. If one new score of 81 is added, what is the new median?

- A) 79.5
- B) 81
- C) 83
- D) 85

TACHS-M6-040

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A regular hexagon has a perimeter of 54 centimeters. What is the length of one side?

- A) 6 centimeters
- B) 8 centimeters
- C) 12 centimeters
- D) 9 centimeters

TACHS-M6-041

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $7 - 2x = -9$.

- A) 8
- B) -1
- C) 1
- D) -8

TACHS-M6-042

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A tank contains 2.75 gallons. If 1 gallon = 4 quarts, how many quarts does it contain?

- A) 9 quarts
- B) 11 quarts
- C) 10 quarts
- D) 12 quarts

TACHS-M6-043

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which expression has a value of -6 ?

- A) $-18 \div 3$
- B) $18 \div (-2)$
- C) $-3 \times (-3)$
- D) $-12 + 4$

TACHS-M6-044

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A fair coin is tossed and a fair six-sided number cube is rolled. What is the probability of tails and an even number?

- A) $\frac{1}{4}$
- B) $\frac{1}{6}$
- C) $\frac{1}{3}$

D) $\frac{1}{2}$

TACHS-M6-045

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A circle has diameter 12 inches. Using $\pi = 3.14$, what is its area?

A) 37.68 in^2

B) 75.36 in^2

C) 113.04 in^2

D) 452.16 in^2

TACHS-M6-046

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

The ordered pairs (1, 5), (2, 8), and (3, 11) follow a linear rule. What is the y-value when $x = 6$?

A) 17

B) 20

C) 18

D) 23

TACHS-M6-047

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A package has a mass of 4.6 kilograms. If 1 kilogram = 1,000 grams, what is its mass in grams?

A) 46 grams

B) 460 grams

C) 4,600 grams

D) 4,060 grams

TACHS-M6-048

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the greatest common factor of 48 and 72?

- A) 24
- B) 8
- C) 12
- D) 18

TACHS-M6-049

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

Set A is 4, 6, 8, 10. Set B is 5, 6, 8, 9. Which statement is true?

- A) Set B has the greater mean and greater range.
- B) Set A has the greater mean and smaller range.
- C) The sets have equal means, but Set A has the greater range.
- D) The sets have equal means and equal ranges.

TACHS-M6-050

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A trapezoid has bases of 8 units and 14 units and height 5 units. What is its area?

- A) 44 square units
- B) 50 square units
- C) 110 square units
- D) 55 square units

SECTION 4 - ABILITY

TACHS-A6-038

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. From left to right, one symbol moves from outside the square to inside it.

Cell 1: + + outside ☐; none inside

Cell 2: + outside ☐; + inside ☐

Cell 3: ?

- A) No + outside ☐; two + inside ☐
- B) Two + outside ☐; one + inside ☐
- C) No + outside ☐; one + inside ☐

D) One + outside □; two + inside □

TACHS-A6-039

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, both arrows rotate 90° clockwise.

Row 1: ↑ ↓ | → ←

Row 2: ← → | ?

A) ↑ ↓

B) ← →

C) ↓ ↑

D) → ←

TACHS-A6-040

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the right cell removes the filled shape and duplicates the open shape.

Row 1: ○ ● | ○ ○

Row 2: □ ■ | ?

A) ■ ■

B) □ □

C) □ ■

D) □ □ □

TACHS-A6-041

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Down each column, the top and bottom symbols combine by placing the bottom symbol inside the top symbol.

Top-left: ○

Bottom-left: +

Result-left: + inside ○

Top-right: □

Bottom-right: ×

Result-right: ?

- A) ☐ inside ×
- B) + inside ☐
- C) × beside ☐
- D) × inside ☐

TACHS-A6-042

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from top to bottom. Two holes are punched near the folded paper's left and right sides at the same height, away from edges. What appears after unfolding?

- A) Four holes in one vertical column
- B) Two holes on the horizontal centerline
- C) Four holes forming the corners of a rectangle
- D) Eight holes around the perimeter

TACHS-A6-043

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangle is folded from left to right. A narrow rectangular notch is cut from the folded paper's top outer edge, away from corners. What appears after unfolding?

- A) One notch at the top center
- B) Four notches, one on each side
- C) Two notches on opposite top and bottom edges
- D) Two matching notches on the top edge, mirrored left-to-right

TACHS-A6-044

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded bottom to top and then right to left. A hole is punched on the folded paper's horizontal outer edge, away from corners and fold lines. What appears after unfolding?

- A) Two holes on the top outer edge
- B) Four holes at the center
- C) Four holes, two on each opposite horizontal edge
- D) Eight holes along every edge

TACHS-A6-045

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded once along the diagonal from lower-left to upper-right. Three holes are punched away from all edges and from one another. What appears after unfolding?

- A) Six holes forming three mirrored pairs across the diagonal
- B) Four holes in two pairs
- C) Nine holes around the center
- D) Three holes on the diagonal

TACHS-A6-046

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

- 1. ○ + □
- 2. △ + ◇
- 3. ☆ + ○

Which choice follows the same rule?

- A) ○ + ●
- B) ■ + □
- C) ◇ × △
- D) □ + △

TACHS-A6-047

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following figures share one rule:

- 1. Outer □; inside ○ at top-left
- 2. Outer ◇; inside △ at top-left
- 3. Outer ○; inside ☆ at top-left

Which choice follows the same rule?

- A) Outer ■; inside ○ at top-left
- B) Outer □; inside ○ at bottom-left
- C) Outer △; inside □ at top-left

D) Outer ; inside  at top-left

TACHS-A6-048

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following sequences share one rule:

1. +  + 

2. +  + 

3. +  + 

Which choice follows the same rule?

A) +  + 

B) +  + 

C)  +  +

D) +  × 

TACHS-A6-049

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. Top ; bottom 

2. Top ; bottom 

3. Top ; bottom 

Which choice follows the same rule?

A) Top ; bottom 

B) Top ; bottom 

C) Top ; bottom 

D) Top ; bottom 

TACHS-A6-050

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share one rule:

1.   

2.   

3. △ △ ▲

Which choice follows the same rule?

A) ◆ ◆ ◇

B) ☆ ★ ☆

C) ○ ● ●

D) ◇ ◇ ◆

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R6-039 - Correct: A - Correct Answer: A sunbeam heats its black plastic case. - Explanation: Direct sunlight warms the sensor's dark case, causing it to register local heat rather than general air temperature.

TACHS-R6-040 - Correct: D - Correct Answer: It helps document the measurement problem they discovered. - Explanation: The marked reading becomes evidence of how placement affected the sensor and supports their explanation.

TACHS-R6-041 - Correct: C - Correct Answer: investigating an unexpected result can deepen a project. - Explanation: The unusual point prompts discussion because the students investigated and explained its cause.

TACHS-R6-042 - Correct: C - Correct Answer: accepted as true without first verifying. - Explanation: Dev initially accepted that the sensor represented greenhouse air before testing that belief.

TACHS-R6-043 - Correct: D - Correct Answer: Their bases attach to flexible skin. - Explanation: The passage explains that scale bases attach to flexible skin, allowing slight movement and overlap.

TACHS-R6-044 - Correct: D - Correct Answer: Pangolin scales combine overlap, curvature, and layers to provide protection while permitting movement. - Explanation: The passage explains several structural features and their combined value for flexible protection and design inspiration.

TACHS-R6-045 - Correct: A - Correct Answer: sent along a different path. - Explanation: Fiber layers can cause a crack to change direction rather than continue straight through.

TACHS-R6-046 - Correct: B - Correct Answer: To emphasize that designers can adapt underlying structural principles. - Explanation: The author distinguishes direct imitation from applying broader ideas such as overlap, curvature, and layered fibers.

TACHS-R6-047 - Correct: A - Correct Answer: Direction - Explanation: The second paragraph states that arrows showed direction.

TACHS-R6-048 - Correct: B - Correct Answer: Excessive detail makes the notation difficult to use. - Explanation: Crowded pages were hard to read, so Maris focused on essential structural decisions.

TACHS-R6-049 - Correct: C - Correct Answer: A selective notation system helps Maris preserve and develop choreography. - Explanation: The revised notation captures useful structure and becomes a tool for reconstruction and composition.

TACHS-R6-050 - Correct: B - Correct Answer: A problem leads to an initial solution, revision, and additional use. - Explanation: Maris responds to imperfect records, simplifies her crowded system, and then uses it as a composing tool.

WRITTEN EXPRESSION ANSWERS

TACHS-WE6-039 - Correct: C - Correct Answer: Sentence 2 - Explanation: Sentence 2 identifies the missing communication that causes later groups to repeat troubleshooting.

TACHS-WE6-040 - Correct: A - Correct Answer: Entries should be specific; "Microphone 4 produces a crackling sound" is more useful than "broken." - Explanation: A semicolon correctly joins the independent clause with the complete example that follows.

TACHS-WE6-041 - Correct: B - Correct Answer: does not function as expected - Explanation: "Does not function as expected" is precise and appropriately formal without claiming the item is completely unusable.

TACHS-WE6-042 - Correct: D - Correct Answer: Students could review the log each month to identify recurring equipment problems. - Explanation: A monthly review logically extends the log's purpose by using accumulated records to identify patterns.

TACHS-WE6-043 - Correct: D - Correct Answer: By day ten, however, both covered plots were moister than an uncovered comparison area. - Explanation: The plural subject "plots" requires the plural past-tense verb "were."

TACHS-WE6-044 - Correct: A - Correct Answer: After receiving equal amounts of water, one plot was covered with five centimeters of straw and the other with five centimeters of wood chips. - Explanation: Choice A clearly combines the controlled watering and equal cover depths without a run-on or misplaced modifier.

TACHS-WE6-045 - Correct: B - Correct Answer: During days 1 through 3, moisture readings decreased more quickly in the straw-covered plot. - Explanation: Choice B specifies the time period, measurement, direction of change, and plot involved.

TACHS-WE6-046 - Correct: A - Correct Answer: Each of the volunteers has received a name tag. - Explanation: The singular subject "Each" correctly takes the verb "has."

TACHS-WE6-047 - Correct: B - Correct Answer: She asked whether the package had arrived. - Explanation: "Whether" and past perfect "had arrived" are used correctly in the reported question.

TACHS-WE6-048 - Correct: C - Correct Answer: Line 3 - Explanation: Line 3 requires the possessive word "their," not the location word "there."

TACHS-WE6-049 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 should use "looks" to maintain the present-tense instructional style.

TACHS-WE6-050 - Correct: A - Correct Answer: Line 3 - Explanation: Line 3 incorrectly uses an apostrophe for a plural; "books" is the correct form.

MATHEMATICS ANSWERS

TACHS-M6-038 - Correct: B - Correct Answer: $5/6$ - Explanation: $2 \frac{2}{3} = 16/6$ and $1 \frac{5}{6} = 11/6$; the difference is $5/6$.

TACHS-M6-039 - Correct: B - Correct Answer: 81 - Explanation: The ordered six scores have 81 and 81 in the middle, so the median is 81.

TACHS-M6-040 - Correct: D - Correct Answer: 9 centimeters - Explanation: A regular hexagon has six equal sides, so $54 \div 6 = 9$ centimeters.

TACHS-M6-041 - Correct: A - Correct Answer: 8 - Explanation: Subtract 7 to get $-2x = -16$, then divide by -2 to get $x = 8$.

TACHS-M6-042 - Correct: B - Correct Answer: 11 quarts - Explanation: $2.75 \times 4 = 11$ quarts.

TACHS-M6-043 - Correct: A - Correct Answer: $-18 \div 3$ - Explanation: $-18 \div 3 = -6$; the other values are -9 , 9 , and -8 .

TACHS-M6-044 - Correct: A - Correct Answer: $1/4$ - Explanation: $P(\text{tails}) \times P(\text{even}) = 1/2 \times 3/6 = 1/4$.

TACHS-M6-045 - Correct: C - Correct Answer: 113.04 in^2 - Explanation: The radius is 6, so area = $3.14 \times 6^2 = 113.04$ square inches.

TACHS-M6-046 - Correct: B - Correct Answer: 20 - Explanation: The y-values increase by 3, so $y = 3x + 2$; when $x = 6$, $y = 20$.

TACHS-M6-047 - Correct: C - Correct Answer: 4,600 grams - Explanation: Multiply 4.6 by 1,000 to obtain 4,600 grams.

TACHS-M6-048 - Correct: A - Correct Answer: 24 - Explanation: The greatest factor shared by 48 and 72 is 24.

TACHS-M6-049 - Correct: C - Correct Answer: The sets have equal means, but Set A has the greater range. - Explanation: Both sets total 28 and have mean 7; Set A's range is 6 while Set B's is 4.

TACHS-M6-050 - Correct: D - Correct Answer: 55 square units - Explanation: Area = $1/2 \times (8 + 14) \times 5 = 55$ square units.

ABILITY ANSWERS

TACHS-A6-038 - Correct: A - Correct Answer: No + outside $\square$; two + inside $\square$ - Explanation: One plus moves inside at each step, so the last cell has both plus signs inside and none outside.

TACHS-A6-039 - Correct: A - Correct Answer: $\uparrow \downarrow$ - Explanation: Rotating $\leftarrow$ clockwise gives $\uparrow$, and rotating $\rightarrow$ clockwise gives $\downarrow$.

TACHS-A6-040 - Correct: B - Correct Answer: $\square \square$ - Explanation: Removing $\blacksquare$ and duplicating $\square$ produces two open squares.

TACHS-A6-041 - Correct: D - Correct Answer: $\times$ inside $\square$ - Explanation: The bottom symbol $\times$ is placed inside the top symbol $\square$.

TACHS-A6-042 - Correct: C - Correct Answer: Four holes forming the corners of a rectangle - Explanation: Each punch reflects vertically, producing two upper and two lower holes arranged as a rectangle.

TACHS-A6-043 - Correct: D - Correct Answer: Two matching notches on the top edge, mirrored left-to-right - Explanation: The vertical unfold mirrors the top-edge notch to a second position on the same top edge.

TACHS-A6-044 - Correct: C - Correct Answer: Four holes, two on each opposite horizontal edge - Explanation: The vertical fold doubles the punch sideways, and the horizontal fold mirrors both to the opposite horizontal edge.

TACHS-A6-045 - Correct: A - Correct Answer: Six holes forming three mirrored pairs across the diagonal - Explanation: Each of three off-fold punches reflects once, producing six holes in three mirrored pairs.

TACHS-A6-046 - Correct: D - Correct Answer: $\square + \triangle$ - Explanation: Two different open shapes appear with a plus sign between them.

TACHS-A6-047 - Correct: C - Correct Answer: Outer $\triangle$; inside $\square$ at top-left - Explanation: An open shape contains a different open shape positioned at the top-left.

TACHS-A6-048 - Correct: A - Correct Answer: $+ \square + \triangle$ - Explanation: The sequence alternates plus signs with two different open shapes, beginning with $+$.

TACHS-A6-049 - Correct: D - Correct Answer: Top $\blacklozenge$; bottom $\circ$ - Explanation: A filled shape is above a different open shape.

TACHS-A6-050 - Correct: D - Correct Answer: $\diamond \diamond \blacklozenge$ - Explanation: Each group has two matching open shapes followed by one filled version of that same shape.